Seasonal Classification and Traffic Forecasting in Mobile Network Cells

Bachelor Project

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Contents

1 Introduction

Mobile networks are under constant pressure to deliver reliable service, while the resources required to do so, including frequency spectrum, infrastructure, and capacity, remain costly. When operators fail to anticipate future traffic demand, the consequences are wasted capacity, degraded user experience, and inefficient infrastructure investment. Beyond immediate performance, accurate traffic forecasting enables operators to make informed decisions about where and when to upgrade network infrastructure, ensuring that capital is directed toward cells where demand genuinely warrants it

Mobile network cells do not follow a single traffic pattern. A common assumption in network planning is that geographic location is a strong determinant of seasonal traffic behaviour, for instance, that recreational areas are inherently seasonal while urban areas are not. These differences are driven by human behavior, such as commuting and recreational activity. As a result, forecasting across all cells is a diverse time-series problem, and it remains an open question whether geographic labels alone are a reliable basis for grouping cells by their underlying traffic behaviour.

Because of this variation, building a robust forecasting model is not straightforward. One approach is to model each cell independently. This preserves local structure but is computationally expensive and can lead to overfitting, especially for cells with limited historical data. Another approach is to train a single global model across all cells. This allows shared representations of traffic patterns and can capture differences between cell types. However, it reduces interpretability and statistical consistency, since the model combines data from sites with different underlying traffic patterns. A third approach is to pre-classify cells based on their dominant traffic behavior, grouping cells with similar temporal structures. This reduces diversity within each group, making traffic patterns more consistent across cells within the same category. In this setting, simpler statistical models such as ARIMA or OLS could remain competitive when applied to more consistent subsets of cells, compared to more advanced machine learning models.

This thesis investigates whether such a data-driven classification can recover, the seasonal grouping that geography is assumed to provide. Specifically, it addresses two research questions: First, can a data-driven statistical method reliably classify mobile network cells into seasonal and non-seasonal categories based on traffic behaviour alone, independent of geographic labels? Second, does training forecasting models on these subgroups yield better predictive performance than a single global model trained across all cells?

The remainder of this work first describes the dataset and preprocessing steps, then introduces a method for classifying cells based on seasonality and traffic behavior, and finally evaluates several forecasting approaches, including OLS, ARIMA, and XGBoost, in terms of forecasting performance.

2 Theory

2.1 Cellular Network Architecture

A cellular network is a wireless communication system that provides mobile devices with access to internet services and Mobile communication. To provide coverage over large geographical areas, network operators deploy base stations at different locations. Each base station serves users within its coverage area and together they can cover wide geographical areas. As shown in Figure 1, multiple base stations can be deployed to form a connected network covering a wide area.

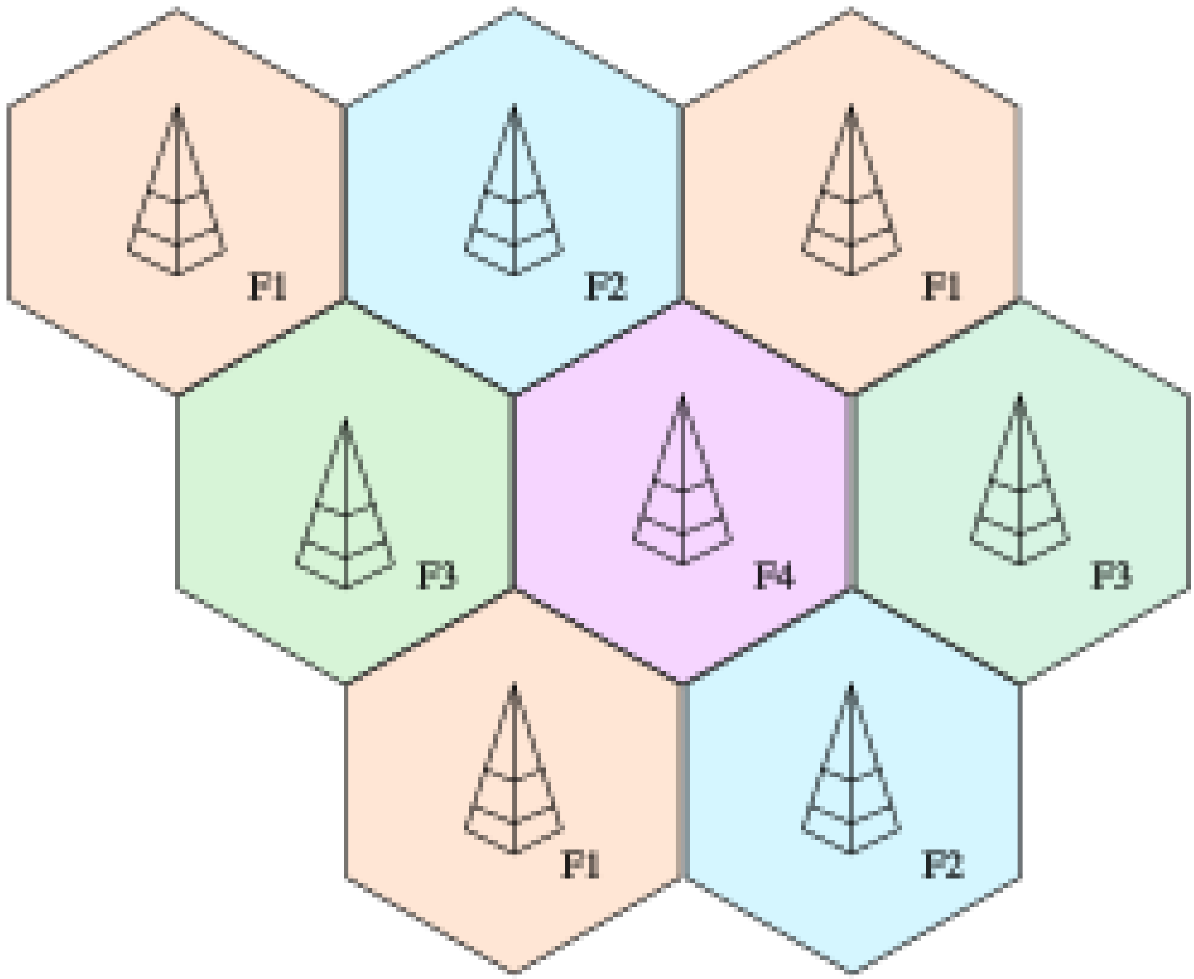


Figure 1: Cellular network architecture showing base stations, user devices, and inter-cell connectivity.

Modern base stations are commonly divided into sectors covering approximately 120 degrees each [1]. Within each sector, multiple cells may operate on different frequency bands and technologies, such as 4G and 5G. A base station consist of multiple cells and the cell is the basic unit of the network and handles communication between user devices and the network.

User devices connect to the network by attaching to the cell that provides the best radio conditions. This decision is mainly based on signal strength and signal quality, which are continuously measured by the device and the network. One key metric is the signal-to-interference-plus-noise ratio (SINR), which describes the quality of the received signal relative to background noise and interference from neighboring cells.

As users move or radio conditions change, connections may be transferred between cells in a process called handover [2]. Handover ensures continuous connectivity by moving the user to a cell with better signal quality or more available capacity when needed.

2.2 Modulation and Signal Quality

Orthogonal Frequency Division Multiplexing (OFDM) is a multi carrier modulation technique that divides a wide frequency band into many narrow, parallel subcarriers [3]. Each subcarrier is shaped as a sinc function in the frequency domain, which produces a peak at its own center frequency and crosses zero at the center frequencies of all neighboring subcarriers as seen in figure 2.

This means that even though the subcarriers overlap in spectrum, sampling each one at its own center frequency yields no contribution from its neighbors, they are mathematically orthogonal. The subcarrier spacing is chosen carefully to be the inverse of the symbol duration, which guarantees these zero crossings align. This allows subcarriers to be packed far more tightly than traditional frequency division multiplexing, where guard bands would be required to prevent interference, resulting in much more efficient use of spectrum.

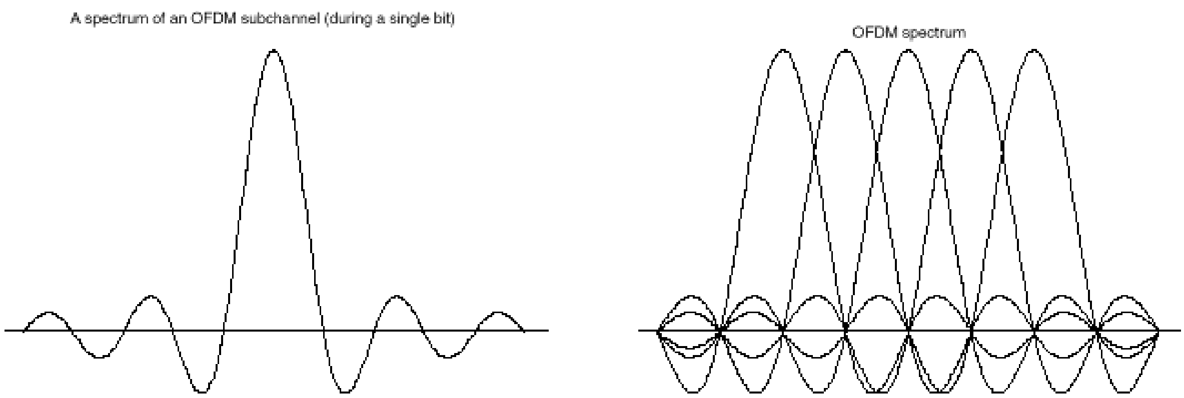


Figure 2: OFDM Sinc function alone vs with 5 subcariers

Modern cellular networks employ adaptive modulation schemes, where the modulation order is selected according to the user's radio conditions [4]. Quadrature Amplitude Modulation (QAM) encodes information by varying both the amplitude and phase of the transmitted signal. Each combination of amplitude and phase corresponds to a unique symbol in the constellation diagram shown in Figure 3.

The number of bits carried by a symbol depends on the modulation order and can be expressed as

$$b = \log_2(M)$$

where M is the number of constellation points. Modulation schemes supported in LTE and NR systems include QPSK, 16QAM, 64QAM, and 256QAM in both uplink and downlink [4], corresponding to 2, 4, 6, and 8 bits per symbol respectively.

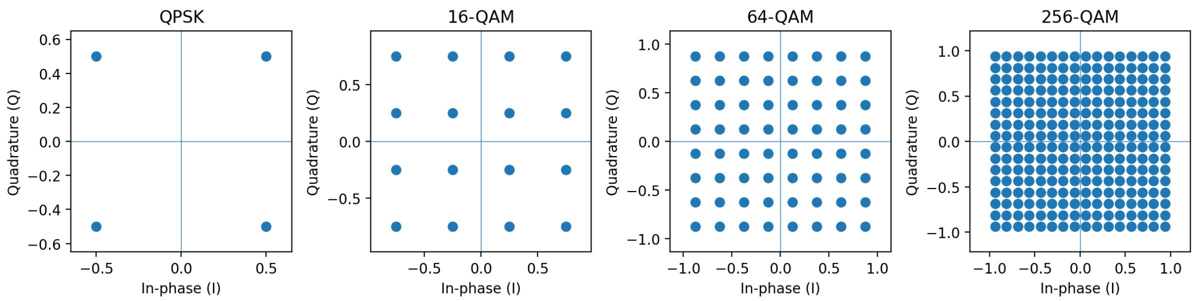


Figure 3: Constellation diagrams for different modulation schemes, illustrating the increasing number of symbols and bits per symbol from QPSK to higher-order QAM.

The modulation order selected by the scheduler depends on the radio conditions experienced by the user equipment, primarily expressed through the Signal-to-Interference-plus-Noise Ratio (SINR) or Channel Quality Indicator (CQI). The base station performs link adaptation based on these measurements, selecting an appropriate Modulation and Coding Scheme (MCS) to match the current channel quality [5]. As channel conditions improve, higher order modulation schemes can be used because constellation points can be placed closer together while remaining reliably distinguishable at the receiver. Conversely, under poor channel conditions, the system falls back to lower order modulation schemes, which are more robust to noise and interference but carry fewer bits per symbol.

The trade-off between modulation order and reliability can be quantified through the Bit Error Rate (BER). For square M-QAM, the theoretical BER as a function of the received E_b/N_0 is well approximated by:

$$\text{BER} \approx \frac{4}{\log_2 M}\left(1 - \frac{1}{\sqrt{M}}\right) \text{Q}\left(\sqrt{\frac{3\log_2 M}{2(M-1)} \cdot \frac{E_b}{N_0}}\right)$$

where M is the modulation order, E_b/N_0 is the energy per bit to noise power spectral density ratio, $\log_2(M)$ is the number of bits carried per symbol, and $Q(x)$ is the standard Gaussian Q-function, representing the probability that a standard normal random variable exceeds x.

Figure 4, higher-order modulation schemes require a significantly larger E_b/N_0 to achieve the same bit error rate, reflecting the reduced distance between constellation points in figure 3

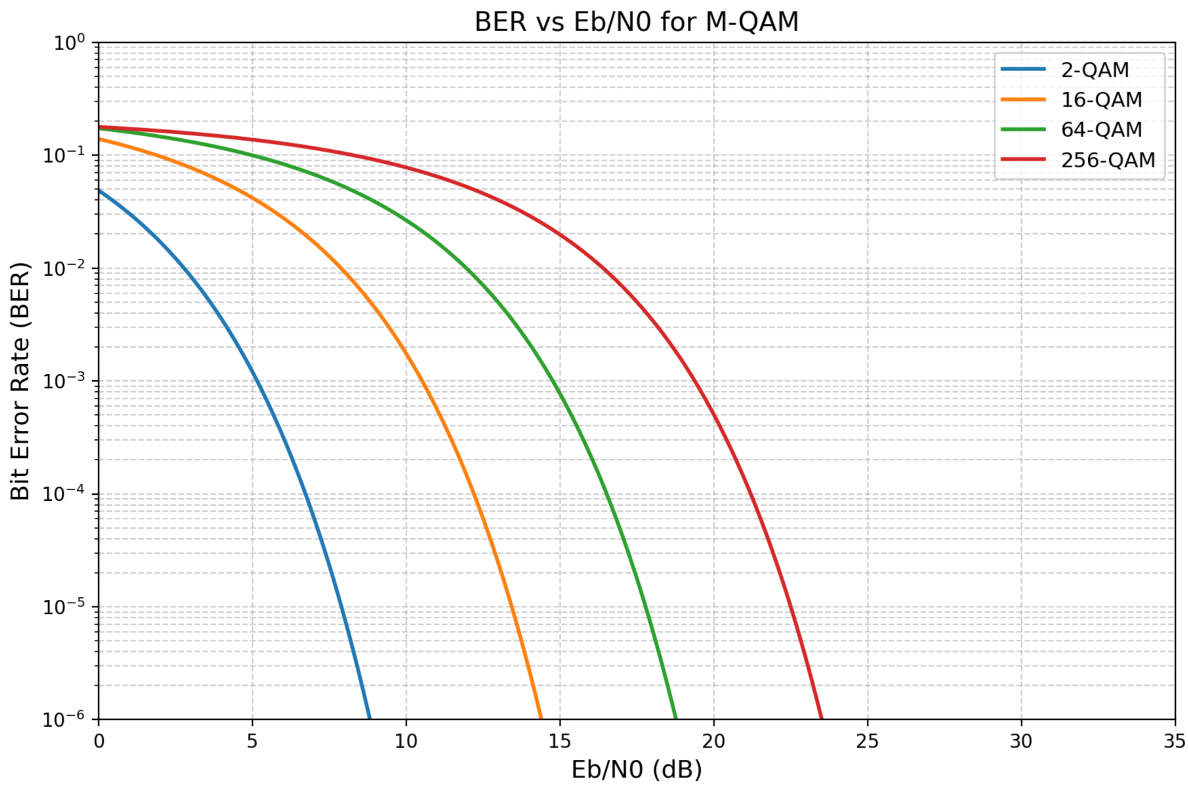


Figure 4: Bit Error Rate plot of different QAM

The result is that two cells with identical PRB utilization can deliver very different data volumes depending on the network conditions experienced by their users, making SINR a fundamental determinant of cell capacity. Beyond its role in per user throughput, SINR is also an important indicator in network planning. Areas with consistently low SINR suggest poor coverage, which may indicate the need for additional infrastructure such as new base stations or upgrades to existing cells. In this way, signal quality metrics inform not only real-time scheduling decisions but also longer-term capacity planning.

2.3 Resource Allocation and PRB Utilization

In 5G NR and 4G LTE, the available frequency spectrum is organized into a two dimensional time frequency grid as seen in figure 5. The frequency axis is 180khz wide divided into subcarriers spaced 15kHz apart [4], while the time axis is divided into symbols. A Physical Resource Block (PRB) consists of 12 consecutive subcarriers in the frequency domain [6]. This grid structure is a direct consequence of the OFDM waveform described in the previous section, where each subcarrier occupies a defined 15khz position in the frequency domain, without any cross talk.

Each cell allocates its available PRBs between connected users through a scheduling process that runs every millisecond. During each scheduling interval, the base station evaluates all active users, their current data demand, and their radio conditions. Based on this information, PRBs are assigned to users according to a scheduling algorithm that balances throughput, fairness, and service requirements.

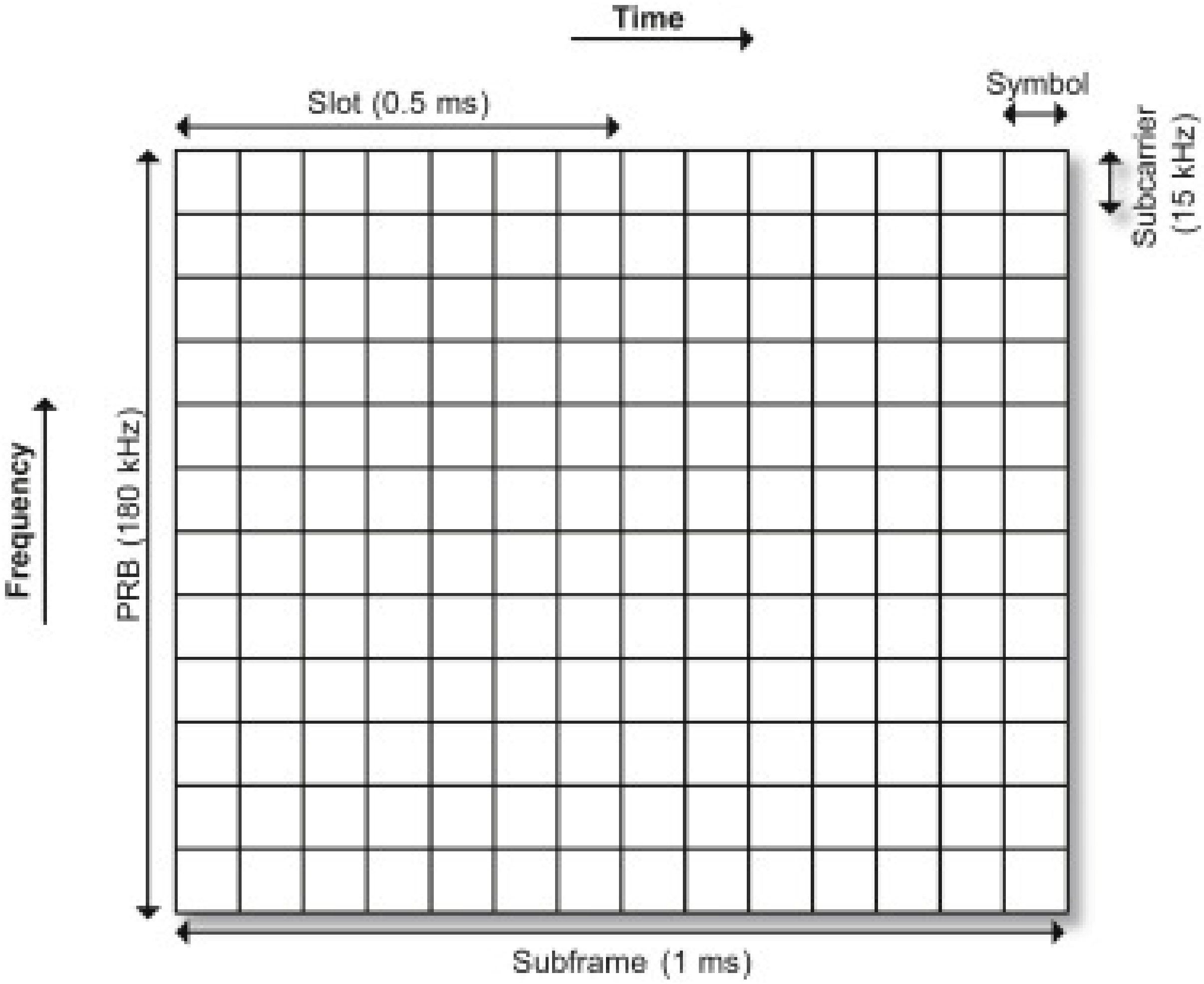


Figure 5: PRB Time and Frequency resource grid

Proportional Fair scheduling

The most commonly used approach is Proportional Fair scheduling algorithm, it aims to achieve a trade off between maximizing total throughput and ensuring fairness among users with different radio conditions [7]. The key idea is to prioritize users who currently have good channel conditions relative to their past achieved throughput.

The PF scheduler computes a scheduling metric for each active user at scheduling time t:

$$M_i(t) = \frac{T_{\text{inst},i}(t)}{[T_{\text{average},i}(t)]^a}$$

where $T_{\text{inst},i}(t)$ is the instantaneous achievable throughput derived from channel quality indicator and the corresponding modulation and coding scheme, and $T_{\text{average},i}(t)$ is the long-term average throughput for user i, obtained using an exponentially weighted low-pass filter. The parameter a controls the fairness–throughput trade-off and is typically set to 1 [7].

Each interval is divided into N time slots, as shown in Figure 5. These time slots are assigned sequentially. After each assignment, the PF metric and throughput averages are updated before the next partition is scheduled. This allows multiple users to be served within a single interval, while still allowing a single user to receive multiple time slots within the same interval depending on their metric.

At each scheduling step, the user with the highest metric is selected for allocation of the current resource unit. This process is repeated over all fixed scheduling resources within the interval rather than resulting in a single user selection for the entire interval.

PRB Utilization

PRB utilization measures the fraction of available resource blocks that are actively used during a reporting interval. It reflects the overall load of the cell by indicating how much of the available capacity is occupied at a given time, independent of how efficiently individual resource blocks are utilised.

High PRB utilization indicates a heavily loaded cell where most available resources are in use, while low utilization indicates available capacity. However, PRB utilization alone does not fully characterize cell performance, since it does not account for the efficiency with which each block is used. The relationship between PRB utilization and throughput provides a more complete picture of cell operating conditions, as illustrated in Figure 6.

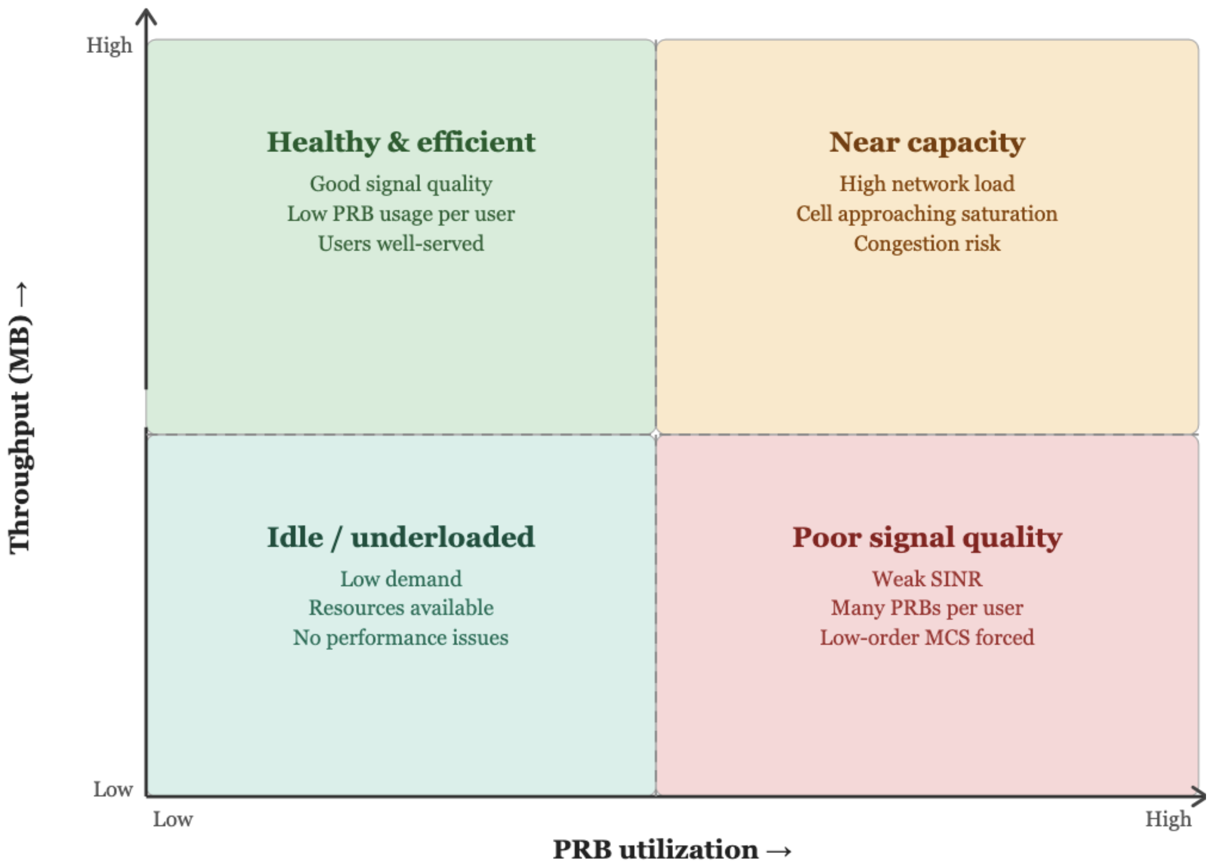


Figure 6: PRB utilization vs Throughput

Four general operating states can be identified from the relationship between throughput and PRB utilization as shown in figure 6. Low PRB utilization combined with low throughput indicates an underloaded cell, where little traffic is being served and most resources remain idle. High PRB utilization combined with low throughput indicates inefficient resource use, typically caused by poor radio conditions that force the scheduler to allocate more PRBs per unit of data. High throughput combined with low PRB utilization indicates efficient operation under favorable radio conditions, where higher-order modulation allows more data to be carried by fewer blocks. Finally, high PRB utilization combined with high throughput indicates a heavily loaded cell operating close to its capacity limit, where resources are being used efficiently but congestion risk increases during periods of peak demand.

Among these states, high PRB utilization combined with low throughput represents the least efficient operating condition. It indicates that the cell is consuming a large share of its available radio resources while delivering relatively little user data, typically as a consequence of degraded signal quality across many connected users.

2.4 Uplink and Downlink Traffic

Traffic in cellular networks is directional. Downlink refers to data transmitted from the network to the user device, while uplink refers to data transmitted from the user device to the network. Applications such as video streaming, web browsing, and social media primarily generate downlink traffic, whereas uplink traffic typically consists of user-generated content, control signalling, and other lower-volume data transmissions. As a result, downlink traffic generally dominates in consumer mobile networks [4].

This asymmetry is reflected in how cellular systems are designed. In Time Division Duplex (TDD) systems, uplink and downlink share the same frequency band but are separated in time [4]. The available resources are divided into time slots that are assigned to either uplink or downlink, allowing the network to adapt to traffic demand. In contrast, Frequency Division Duplex (FDD) systems use separate frequency bands for uplink and downlink, enabling simultaneous transmission in both directions.

In both cases, the overall resource allocation is typically biased heavily toward downlink traffic to reflect the dominant direction of data usage in mobile networks.

2.5 Traffic Generation in Cellular Networks

Cellular network traffic originates from individual user sessions, where a session referes to any period of data exchange between a users device and the network cell. These session can vary widely in terms of duration, data volume, uplink/downlink depeding on the application. Streaming videos generates a sustained high volume downlink session, while messaging generates short burst of low volume traffic, and audio calls generates a sustained low volume traffic.

At any given time, the traffic observed at the cell level is the sum of all active user sessions within that cell. This means that cell level traffic reflects the combined behavior of many independent users rather than a single transmission pattern. As a result, traffic at the network level follows human activity patterns. Users do not generate sessions uniformly over time, but instead concentrate activity around daily routines such as working hours and commuting. This leads to clear daily patterns, with high load during the day and lower activity during the night. Longer-term variations can also appear, including weekly differences between weekdays and weekends, as well as seasonal effects driven by travel and population movement.

Different geographical environments produce structurally distinct traffic patterns. Cells located near event venues such as stadiums and exhibition centers may experience long periods of low activity interrupted by short, sharp peaks triggered by external events. These series are non-stationary, since their statistical properties change abruptly during event windows.

Recreational cells serving destinations such as ski resorts, coastal regions, and holiday areas follow smoother and more regular seasonal cycles. These patterns are visible across months or years, with recurring peaks during specific periods. Traffic in these cells is typically not stable week to week, but exhibits seasonal cycles that repeat year to year.

Urban and commercial cells tend to exhibit a more stable traffic structure. Traffic is driven by regular human activity patterns such as work commutes and daily routines, producing strong and consistent daily and weekly cycles, including differences between weekdays and weekends. These series tend to be closer to stationarity compared to recreational or event-driven cells.

This user activity forms the basis of the measurements collected by the network. The base station continuously records and summarizes these events into cell level indicators, which are then reported as Performance Management (PM) counters at fixed time intervals.

2.6 PM Counters

Mobile network operators typically collect standardized measurements from their network cells using Performance Management (PM) counters. These are software counters embedded in the base station that continuously record network activity and report values at fixed time intervals, typically every 15 minutes or every hour, depending on the network configuration.

PM counters operate at the cell level, meaning that each counter value represents the combined activity of all users connected to a specific cell during the reporting interval. Typical counter categories include traffic volume counters, PRB utilization counters, unique user counters, and signal-to-noise ratio counters. Traffic measurements include both downlink and uplink counters, and most categories are reported as average and maximum values over the interval.

Traffic volume counters are typically implemented as cumulative sums of transmitted and received data bytes at the base station. These counters are incremented based on the actual number of successfully transmitted bits at the MAC or PDCP layer during scheduling. At the end of each reporting interval, the accumulated values are exported and reset or reported as deltas, and higher-level KPIs such as throughput are derived from these measurements.

3 Dataset and Preprocessing

The dataset used in this study is sourced from TDC NET as a part of a collaboration in connection with this bachelor's thesis. It consists of hourly time-series measurements collected from 12 cellular network towers across Denmark. The towers are grouped into three categories, urban, recreational, and suburban. The labels are based solely on the geographical location of each tower. Overall summary of the dataset is shown in table 1.

Table 1: Dataset before preprocessing

Statistic	**After**
Rows	3,104,976
Urban sites (cells)	4 (83)
Suburban sites (cells)	4 (93)
Recreational sites (cells)	4 (71)
Cells total	255
Missing rate (data volume dl)	33.2%
Start date	2024-05-21
End date	2026-03-01

The dataset is structured as a multivariate time series. Each row corresponds to an hourly observation, while each column represents a performance indicator describing the network traffic during that hour. Several features are further separated by traffic direction (uplink and downlink). An overview of the features is provided in Table 2.
To provide a clearer visualization of the dataset, Figure 7 shows the daily peak downlink data volume across 9 different cells: 3 recreational (R), 3 urban (U), and 3 suburban (S) sites. The figure also reveals a gap in August 2024 due to missing data in the dataset.

Already at this stage, clear differences in traffic behavior can be observed. Recreational sites generally show large month-to-month shifts in traffic, indicating high variance, as seen most clearly in R1 2F, R2 1F, and R3 1F. R3 3F is an exception within this category, showing generally low and stable traffic throughout the year with a sudden, sharp spike in February. Urban and suburban sites generally appear more similar to each other in traffic pattern, with most cells showing stable traffic throughout the year. The main exception is U2 1F, which instead shows a steady increase in traffic throughout most of the year, peaking before declining toward the winter months, a pattern more characteristic of the recreational sites than the otherwise stable urban and suburban behavior.

Feature	Type
Signal-to-noise ratio (SINR max)	continuous
Signal-to-noise ratio (SINR avg)	continuous
Peak average users	continuous
Average users	continuous
Peak data volume Downlink	continuous
Peak data volume Uplink	continuous
Average data volume Downlink	continuous
Average data volume Uplink	continuous
PRB utilization Downlink	continous
PRB utilization Uplink	continuous
Average user throughput Downlink	continuous
Average user throughput Uplink	continuous
Site ID	categorical identifier
Cell ID	categorical identifier
Frequency technology (4G/5G)	categorical
Frequency spectrum	categorical

Table 2: Overview of dataset features and their types

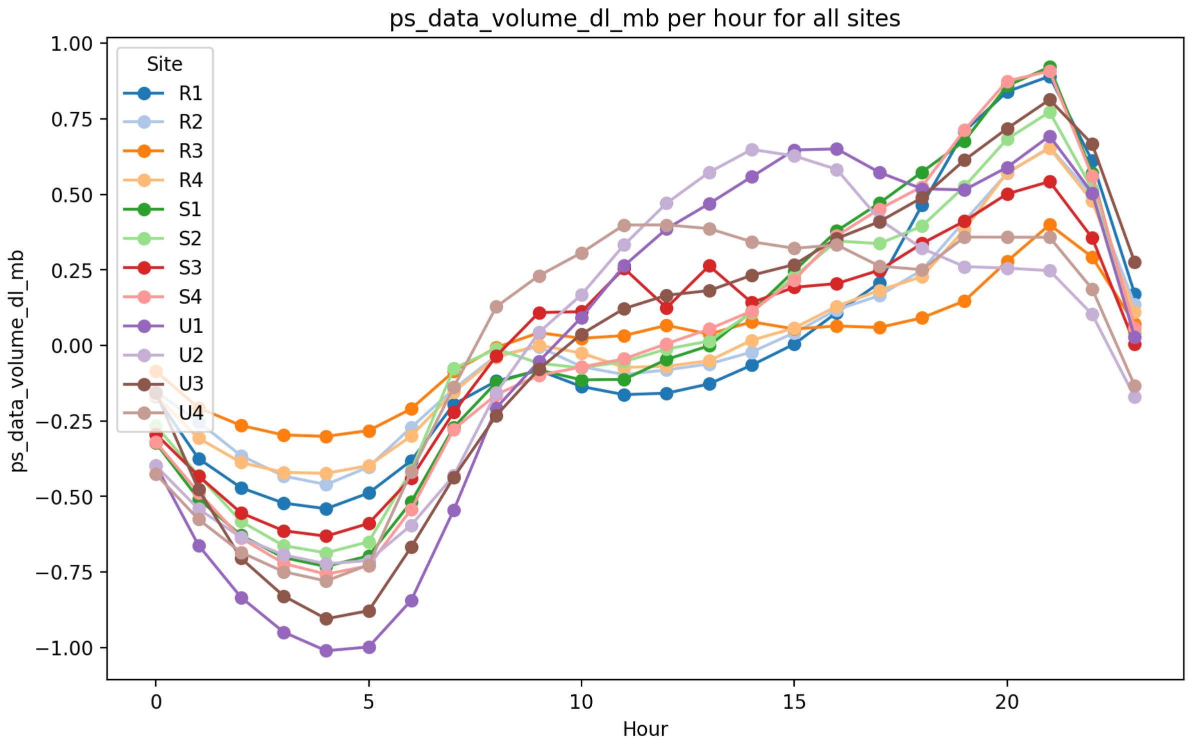


Figure 8: Average traffic volume per hour of day across all cells

Figure 8 shows the average hourly downlink data volume for each site, normalized using z-score standardization, to get a clear comparison in patterns across the sites. Clear differences can be seen between urban, suburban, and recreational cells.

All site types reach their minimum traffic between 03:00 and 05:00, with urban cells (U1–U4) showing the lowest values, with z-scores around (-0.75 to -1). This suggests that

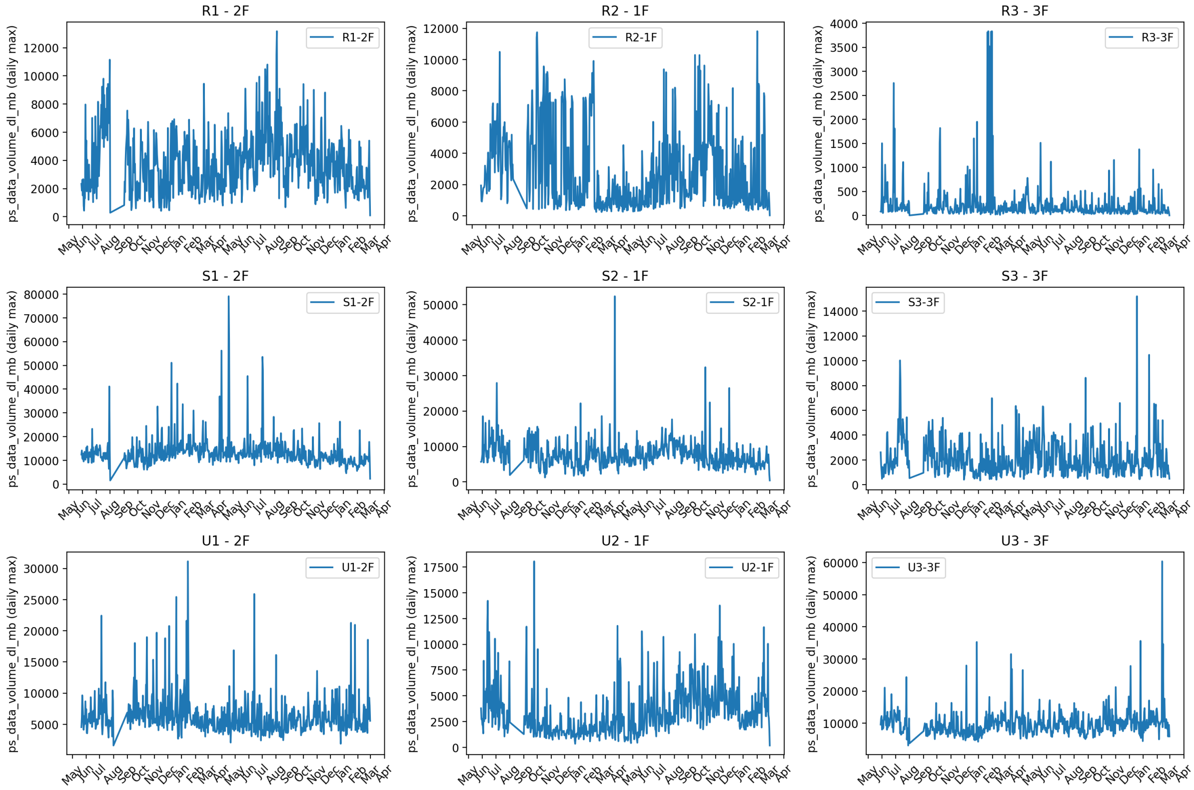


Figure 7: 2F cell Daily peak data volume downlink across all sites

urban cells experience the strongest night-time drop relative to their average level, reflecting a clear difference between peak and off-peak demand in densely populated areas..

Urban cells (U1-U4) recover quickly from the overnight minimum, reaching a mid day peaking around 11:00-16:00 at z-scores of approximately 0.5-0.65, and declining again after 16. Recreational cells (R1-R4) follow a different daily cycle, remaining relatively flat during midday near their mean (z-score ≈ -0.1 – 0.1), before steadily increasing toward an evening peak around 20:00–21:00. Suburban cells (S1–S4) closely mirror this recreational pattern, with low daytime activity and a steady climb toward an evening peak, suggesting that both categories are driven by similar leisure and residential usage behaviour rather than the workplace and commuter patterns that dominate urban cells.

The most striking divergence occurs in the evening. Recreational and suburban cells reach their daily peak between 19:00 and 21:00, with several sites such as R1, S4, and S1 exceeding a z-score of 0.75. Urban cells also rise in the evening but peak slightly earlier, around 20:00, and at comparatively lower normalized levels. This represents an evening peak roughly 30–40% higher in normalized terms than the midday plateau observed in urban cells, highlighting a fundamentally different daily usage cycle.

Notably, U3 and R2 deviate from the dominant patterns within their respective categories. U3 follows a pattern more similar to recreational sites, with low activity during the day and a pronounced evening peak around 20:00.

These results confirm that the timing of peak demand differs systematically by cell type.

Urban cells follow a midday-dominated pattern consistent with workplace and commuter activity, while recreational and suburban cells follow an evening-dominated pattern more consistent with leisure usage. This distinction has direct implications for forecasting, as models trained on mixed cell types risk combining two structurally different daily cycles.

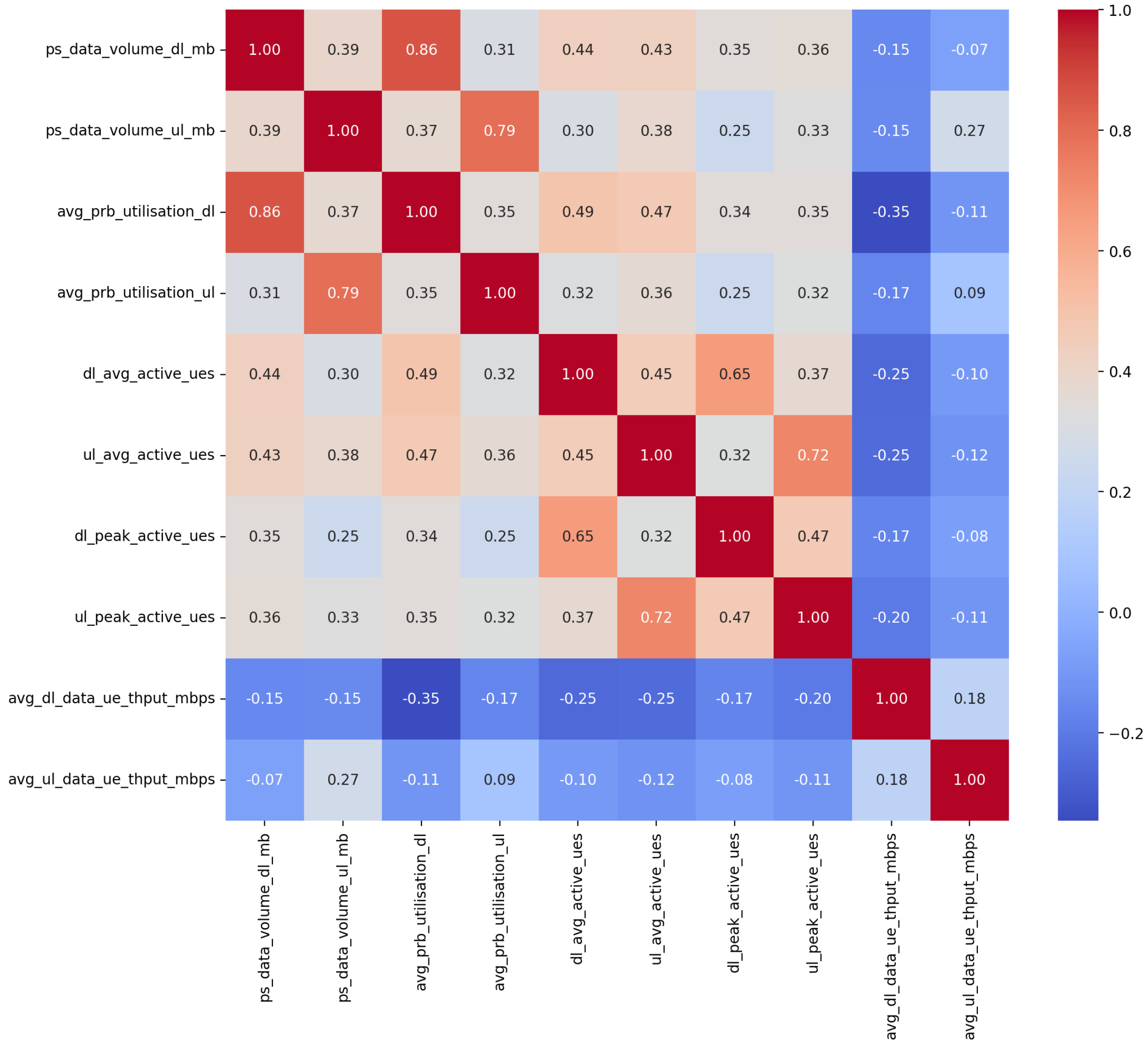


Figure 9: Feature correlation matrix

Figure 9 shows the correlation matrix of all continuous features in the dataset. Most features are positively correlated with each other. The main exception is user throughput (Mbps) negative correlations compared to the other features.

This behavior is expected. When a cell has fewer active users, it can allocate more throughput per user, which can lead to higher throughput values per user when overall traffic is low.

Data volume UL/DL and average PRB utilisation UL/DL are highly correlated at 0.86 for downlink and 0.79 for uplink. This is consistent with the theory presented in Section 2.3, more PRBs allocated generally means more data being transmitted. However, the

correlation is not perfect, as poor radio conditions can force the scheduler to allocate more PRBs per unit of data, as illustrated in Figure 6.
In general, features within the same category show strong correlations, such as average and peak values of the same feature. Another strongly correlated feature is the relationship between PS data volume and average PRB utilization, where higher traffic volume leads to higher resource usage in the cell.

Overall, the results suggest that traffic related metrics are strongly interdependent, meaning that a few key features, such as data volume, already provide a good representation of overall cell.

3.1 Preprocessing

Preprocessing is a crucial step in the analysis, particularly when the goal is to capture seasonal patterns in the data. To reliably observe seasonality, each time series must cover at least one full yearly cycle. Shorter time series may not contain enough information to capture these effects.

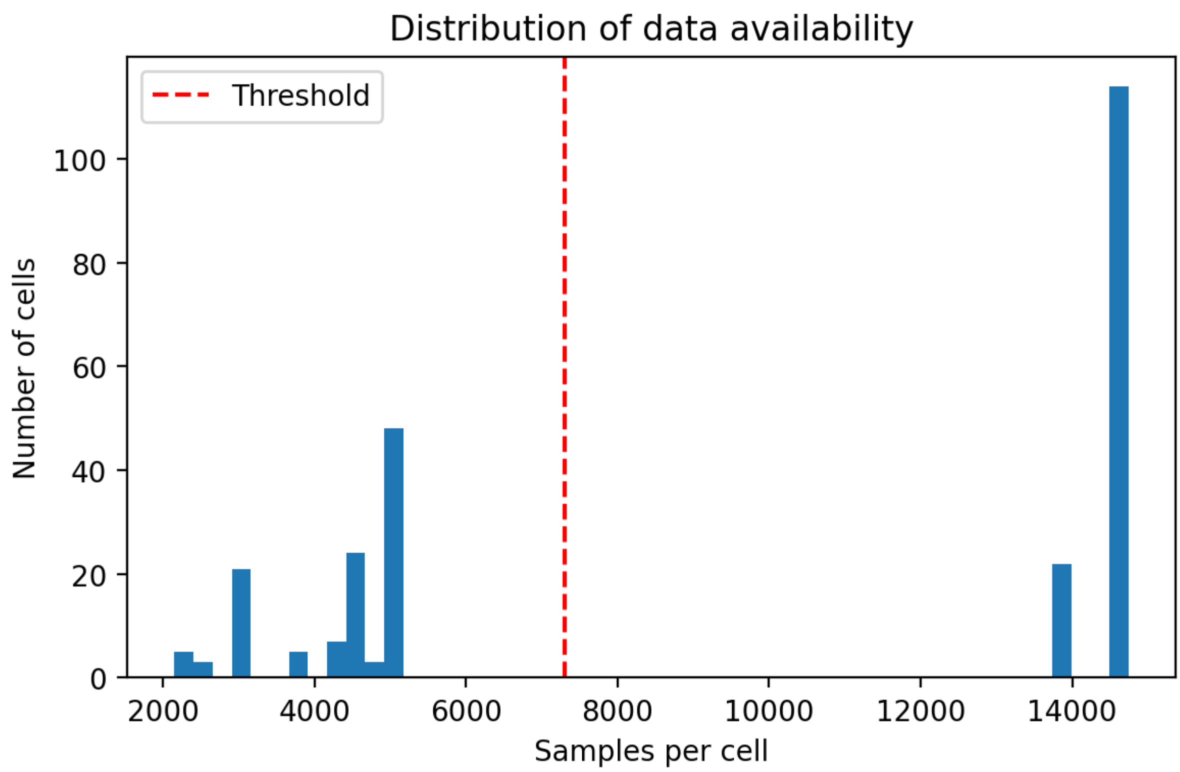


Figure 10: Distribution of available data samples per cell

Figure 10 shows the distribution of the number of observations per cell. Two distinct groups can be observed. The first group contains cells with approximately 15.000 samples, corresponding to more than one year of data. The second group contains cells with roughly 2000-5000 samples, which is below one year of observations.

This resulted in the removal of 125 cells due to insufficient length. The removals were distributed across site types as follows: Urban (44), Suburban (48), Recreational (33).

Two datasets were provided as part of this thesis, one containing raw KPIs directly from the PM counters, and a second containing cell-level metadata, including technology generation and frequency spectrum. During preprocessing, cells present in the KPI dataset but missing from the metadata dataset were dropped as part of the merge step. This affected 29 cells in total: 8 urban, 12 suburban, and 9 recreational.

Duplicate entries at the (site, cell, datetime) level were removed. When duplicates contained conflicting values, rows with non-missing values in the ps data volume dl column were prioritized. This resulted in a deletion of 1,130,288 rows and the dataset going from 3,104,976 to 1,974,688 rows.

Table 3: Dataset after preprocessing

Statistic	**After**
Rows	1,770,806
Urban sites (cells)	4 (52)
Suburban sites (cells)	4 (45)
Recreational sites (cells)	4 (39)
Cells total	136
Missing rate (data volume dl)	166 (0.01%)
Start date	2024-05-21
End date	2026-03-01

Z-score normalization

Since the analysis focuses on patterns across cells rather than absolute values, Z-score normalization is applied to each cell individually. This transformation ensures that each time series has a mean of 0 and a standard deviation of 1.

$$z = \frac{x - \mu}{\sigma} \tag{1}$$

Here, μ is the mean and σ is the standard deviation of the values within each cell.

By applying this normalization, differences in scale between cells are removed, allowing for direct comparison of patterns in traffic load rather than magnitude differences.

4 Network Site Classification

Several approaches can be used for classifying time series data. Unsupervised learning methods such as k-means clustering are commonly used when the structure of the data is unknown. These methods can capture complex patterns but are often less interpretable.

Alternatively, statistical approaches based on hypothesis testing provide more transparent decision rules but may rely on stronger assumptions about the data.

In this study, the objective is to separate cells into two classes, seasonal and non-seasonal. Given the binary nature of the problem and the need for interpretability, a simple and explainable approach is preferred over more complex clustering methods.

To identify a simple and interpretable seasonal indicator, a binary feature based on public school holiday schedules in Copenhagen [8] was constructed. Each time series was labeled as either occurring during a holiday period or a non-holiday period.

For each cell, the traffic data was divided into two groups corresponding to holiday and non-holiday periods. The average traffic volume was then computed for both groups. To obtain an overall overview at the site level, the average values across all cells within each site were visualized using boxplots, as shown in Figure 11.

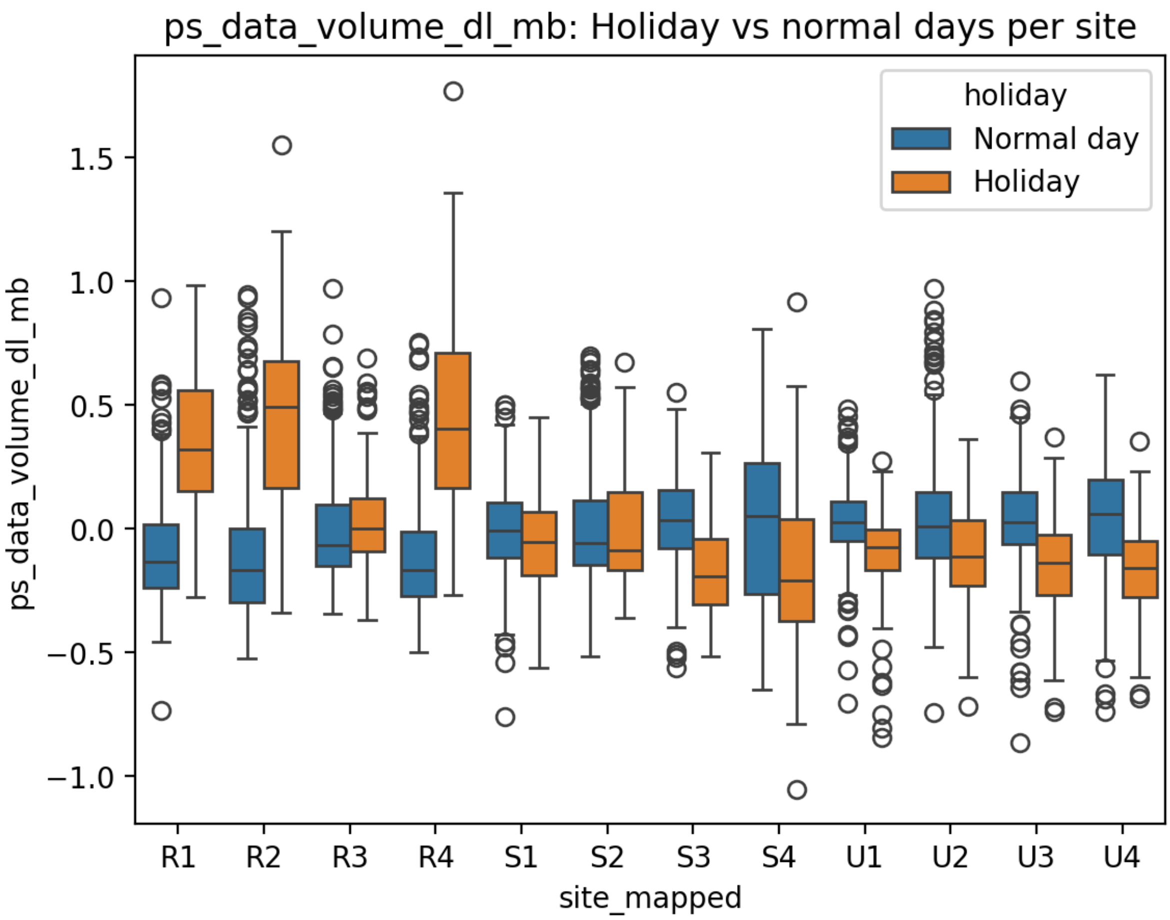


Figure 11: T-test holiday volume vs non-holiday volume for all individual cells

Figure 11 shows differences in standardized traffic levels between holiday and non-holiday periods across site categories [9].
Recreational sites (R1–R4) show a consistent positive shift in standardized traffic during holiday periods. The mean difference between holiday and non-holiday periods

ranges from 0.04 to 0.58 standard deviations. The strongest effects are observed at R4 ($\Delta = 0.582$) and R2 ($\Delta = 0.547$), followed by R1 ($\Delta = 0.442$). R3 shows only a weak shift ($\Delta = 0.051$), indicating a much smaller seasonal response compared to the other recreational sites.

Urban sites (U1–U4) show a consistent negative shift during holiday periods. Mean differences range from -0.13 to -0.20 standard deviations, with the strongest decreases observed at U4 ($\Delta = -0.202$) and U2 ($\Delta = -0.192$). This indicates lower standardized traffic levels during holidays compared to normal days across all urban locations.

Suburban sites (S1–S4) show mixed and weaker effects. S1 ($\Delta = -0.039$) and S2 ($\Delta = -0.020$) show near-zero changes, indicating little difference between holiday and non-holiday periods. S3 ($\Delta = -0.213$) and S4 ($\Delta = -0.141$) show larger negative shifts, more similar to urban patterns. Overall, the suburban group shows higher diversity, with no consistent directional response across all sites.

Median differences follow a similar pattern, with recreational sites showing positive shifts and urban sites showing negative shifts, confirming that the observed effects are not driven solely by extreme values.

4.1 T-test Validation

The boxplot analysis demonstrates that holiday periods produce different traffic volumes depending on site category, suggesting that the holiday indicator can serve as a basis for statistical classification at the cell level. To formalize this, an independent two-sample t-test is applied to each cell individually, testing whether the mean traffic volume during holiday periods differs significantly from that during non-holiday periods.

The t-statistic is defined as:

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} \tag{2}$$

where $\bar{x}_1$ and $\bar{x}_2$ are the sample means, s_1^2 and s_2^2 are the sample variances, and n_1 and n_2 are the number of observations in the holiday and non-holiday groups.

The t-test is not used here as a classifier in the traditional sense, but rather as a statistical decision rule. The underlying hypothesis is that cells with seasonal traffic patterns will respond differently to holiday periods than non-seasonal cells. The two groups are naturally defined and non-overlapping, and the test provides a transparent statistical measure of whether the observed difference in mean traffic is likely to occur by chance. Cells with statistically significant differences ($p < 0.05$) are labeled as seasonal, while the remaining cells are labeled as non-seasonal. This produces an interpretable and reproducible classification criterion without requiring manual threshold tuning.
Figure 12 shows the t-test results across all individual cells, and Table 4 summarizes the resulting classifications by site type.
A strong separation is observed for urban and recreational sites. All 52 urban cells are classified as non-seasonal (0 seasonal, 52 non-seasonal), which aligns with the expecta-

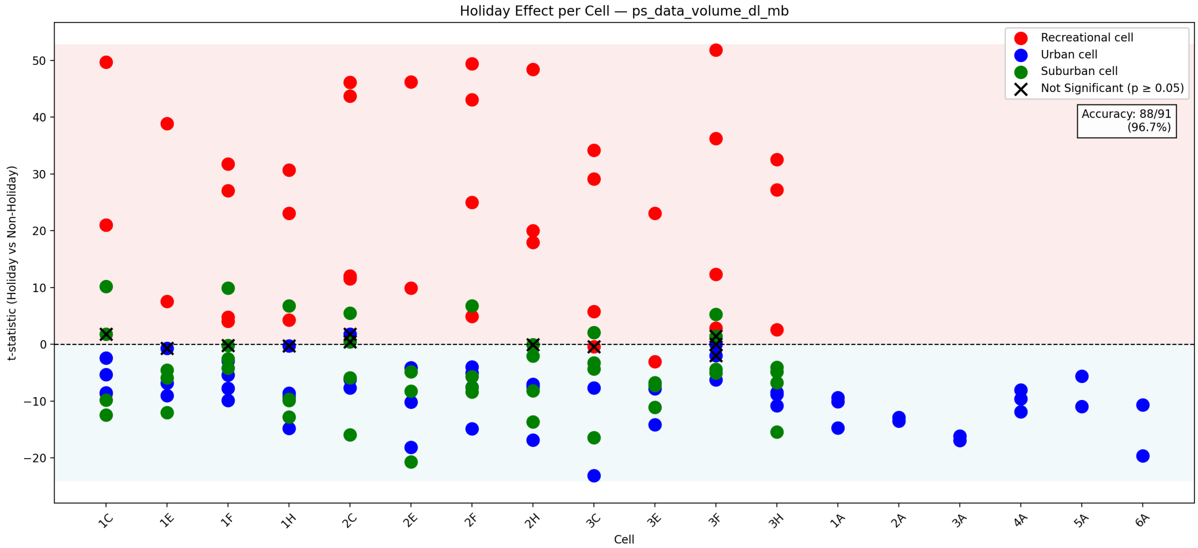


Figure 12: T-test holiday volume vs non-holiday volume for all individual cells

Table 4: Seasonal vs non-seasonal counts by site type

	Seasonal	Non-seasonal
Recreational (R)	36	3
Suburban (S)	7	38
Urban (U)	0	52

tion that urban traffic is stable across holiday and non-holiday periods. Recreational sites show the opposite pattern, with 36 of 39 cells classified as non-seasonal and 3 classified as seasonal.

The suburban category is more mixed. Out of 45 suburban cells, 38 are classified as seasonal and 7 as non-seasonal. This split reflects the diverse nature of suburban areas, where traffic patterns are more in the middle of seasonal and non-seasonal.

Overall, the results indicate strong agreement between the method and expected behavior for urban and recreational areas, while suburban areas show a more mixed classifications. Across urban and recreational sites, 88 out of 91 cells are consistent with their expected class based on geographic category. All urban cells (52/52) are classified as non-seasonal, while recreational sites show 36 of 39 cells classified as non-seasonal. This level of agreement provides supporting evidence for the method, even in the absence of ground truth labels at the individual cell level, and highlights the value of a data driven approach that does not rely on geographic labels.

Given the strong separation achieved by this simple method, the focus of the remaining work shifts to evaluating whether this classification actually improves forecasting performance, rather than exploring more complex classification alternatives such as k-means clustering, dynamic time warping similarity, or supervised classification methods.

5 Traffic Forecasting

This section builds on the classification results presented earlier. The motivation for classification is to better understand underlying traffic patterns before performing forecasting. By separating network sites into seasonal and non-seasonal groups, it becomes possible to evaluate whether different traffic dynamics require different forecasting strategies.

All forecasting models are evaluated using a chronological 80/20 train-test split. The first 80% of each time series is used for training, while the final 20% is reserved for testing. This split preserves temporal order and avoids information leakage from future observations. The test set is used as the common validation set across all models.

Root Mean Squared Error (RMSE)
Model performance is evaluated using the Root Mean Squared Error (RMSE), defined as:

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \tag{3}$$

where y_i is the observed value and $\hat{y}_i$ is the predicted value. RMSE penalizes larger errors more heavily than absolute error due to the squaring term, while remaining in the same unit as the target variable. This makes it suitable for network traffic forecasting, where large underestimations of peak load can lead to capacity issues.

5.1 Baseline Models

Before introducing more complex forecasting approaches, several baseline models are defined to provide reference performance levels [10]. These baselines allow for a fair evaluation of whether more advanced methods provide meaningful improvements.

The dataset is evaluated under three settings, seasonal cells, non-seasonal cells, and a combined dataset. This enables analysis of whether traffic behavior differs across network environments. The cells were split into the different groups by the t-test as shown in figure 2, if the p-value was below 0.05, there was not enough evidence that the cell was seasonal and it was put in the non-seasonal category.

The forecasting target is the daily peak traffic volume, defined as the maximum observed hourly traffic within a day as seen in figure 7. This metric is particularly important for network planning, as it reflects peak load conditions that are critical for capacity management.

Table 5 presents the baseline model results. The seasonal naive model performs worst across all splits, with an RMSE of 2.36 on seasonal cells, 2.58 on non-seasonal cells, and 2.51 overall. This suggests that traffic patterns are not stable year over year.

Model	Seasonal RMSE	Non-Seasonal RMSE	Overall RMSE
Seasonal Naive	2.36	2.58	2.51
Mean Naive	1.85	1.93	1.93
Previous Naive	1.77	1.99	1.92

Table 5: RMSE obtained by the baseline forecasting models using the 80/20 chronological train-test split. Lower values indicate better forecasting performance.

A notable pattern emerges when comparing the mean and previous value models across dataset splits: the mean naive model performs better on non-seasonal cells (1.93 vs. 1.99), while the previous value model performs better on seasonal cells (1.77 vs. 1.85). This is consistent with the underlying traffic patterns, as non-seasonal cells show relatively stable behavior over time, making the mean a reliable predictor, whereas seasonal cells are subject to gradual trends and periodic fluctuations, meaning recent observations carry more predictive information than the historical average.

It should be noted that RMSE values are not directly comparable across dataset splits, as seasonal and non-seasonal cells differ in their underlying traffic variance. Comparisons are therefore made within each group independently.

5.2 ARIMA

ARIMA (AutoRegressive Integrated Moving Average) is a widely used statistical model for forecasting time series data. It is designed to capture patterns such as autocorrelation and trends in the data.
The model is defined by three parameters: p, d, and q.

- p: the number of autoregressive (AR) terms, which model the relationship between the current value and its past values.
- d: the degree of differencing used to make the series stationary.
- q: the number of moving average (MA) terms, which model the relationship between the current value and past forecast errors.

A seasonal extension (P, D, Q, m) can also be included, where m represents the number of time steps in one seasonal cycle. This allows the model to capture repeating patterns that occur every m periods.

The optimal parameters are selected by minimizing the Akaike Information Criterion (AIC), which balances model fit and complexity. [11]

SARIMAX

SARIMAX is an extension of the ARIMA model. The S represents the seasonal component introduced previously, while the X indicates the inclusion of exogenous input features. Unlike ARIMA, which only models the historical behavior of the time series itself, SARIMAX can additionally incorporate external variables that may influence the forecasting.

Model Optimization

The following method was used to validate the ARIMA forecasting approach on the dataset. As described earlier, the data consists of two groups seasonal and non-seasonal. For each individual cell time series, the auto ARIMA function [10] was applied to determine the parameter set that minimizes the AIC. The most frequently selected order across all cells was then chosen as the representative seasonal and non-seasonal configuration.

The proposed evaluation procedure consisted of three stages:

1. First, the baseline ARIMA configuration was determined using auto-ARIMA without any exogenous variables.
2. Second, forward feature selection was applied to the exogenous variables using the fixed baseline ARIMA order.
3. Finally, after selecting the optimal feature subset, the ARIMA coefficients and model parameters were re-estimated using the selected exogenous variables to obtain the final SARIMAX model

baseline ARIMA configuration

Figure 13 shows the frequency of the ARIMA orders selected by the auto-ARIMA procedure across all seasonal cell time series.

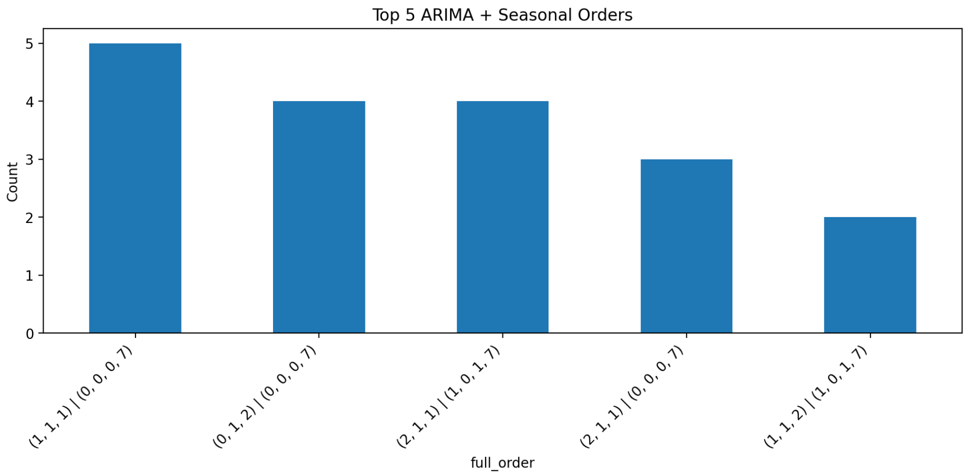
For the seasonal setting, the most frequently selected configuration was $\mathrm{ARIMA}(1,1,1)(0,0,0,7)$. This indicates that first-order differencing was consistently required, while the dominant structure was captured through short-term autoregressive and moving-average components.

For the non-seasonal setting, the dominant configuration was $\mathrm{ARIMA}(0,1,1)(0,0,0,7)$. Compared to the seasonal case, the autoregressive component disappeared, suggesting weaker temporal dependence in the non-seasonal cells. However, first-order differencing, the moving-average and seasonal components remained consistently selected across both groups.

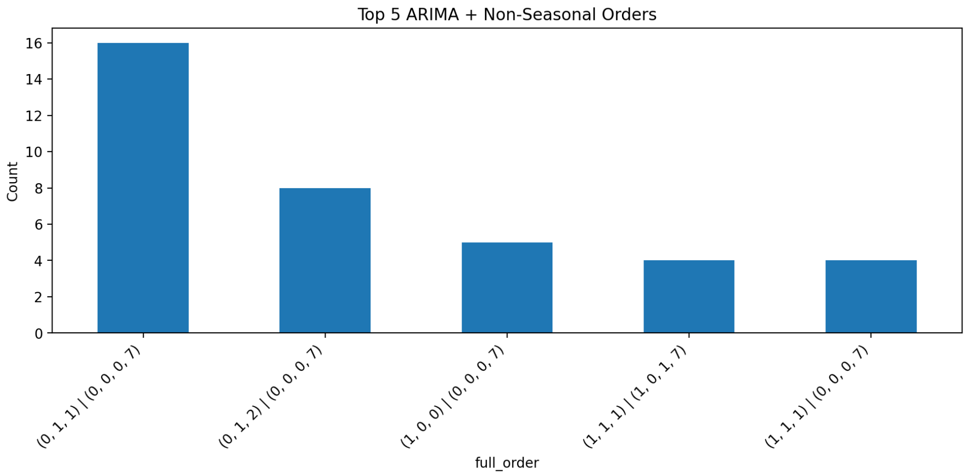
Feature selection

To evaluate the impact of exogenous variables on forecasting performance, a forward feature selection approach was used. The goal was to iteratively build a feature set that minimizes the mean squared error (MSE) across all time series (cells).

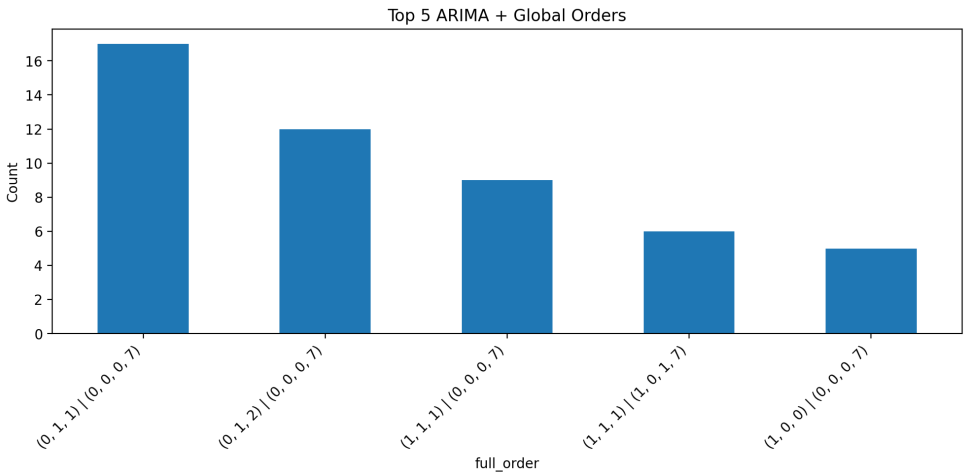
Table 6 shows the set of candidate features evaluated during the selection process.
To evaluate the impact of exogenous variables on forecasting performance, a forward feature selection approach is applied [12]. The method starts with an empty feature set and iteratively adds the feature that yields the largest reduction in average test root mean squared error (RMSE) across all time series. Candidate features include calendar



(a) ARIMA order frequency for seasonal cell
s



(b) ARIMA order frequency for non-seasonal cell



(c) ARIMA order frequency for all cells

Figure 13: Arima order before feature selection

Table 6: Candidate features evaluated during feature selection

Feature	**Description**
month	Month of the year
dayofweek	Day of the week
is_weekend	Weekend indicator variable
holiday	Public holiday indicator
lag1	Previous day traffic value
lag7	Traffic value from 7 days earlier
roll_mean_7	Rolling 7-day mean
roll_mean_14	Rolling 14-day mean
roll_std_7	Rolling 7-day standard deviation
roll_std_14	Rolling 14-day standard deviation
roll_min_7	Rolling 7-day minimum
roll_max_7	Rolling 7-day maximum
lag_diff_1_7	Difference between lag-1 and lag-7

variables, lag-based features, rolling statistics, and differenced features. The process continues until no further improvement is observed, as shown in Algorithm 1.

Algorithm 1 Forward Feature Selection for ARIMA with Exogenous features

Initialize selected feature set $S \leftarrow \emptyset$
Initialize remaining features $R \leftarrow \{f_1, f_2, ..., f_n\}$
Set best error $E_{best} \leftarrow \infty$
while $R \neq \emptyset$ **do**
    $f^* \leftarrow \emptyset$
    $E^* \leftarrow E_{best}$
    for each feature $f \in R$ **do**
        Train ARIMA models with feature set $S \cup \{f\}$ across all series
        Evaluate mean test error E_f
        if $E_f < E^*$ **then**
            $E^* \leftarrow E_f$
            $f^* \leftarrow f$
        end if
    end for
    if $f^* \neq \emptyset$ **then**
        Add f^* to S
        Remove f^* from R
        Update $E_{best} \leftarrow E^*$
    else
        Stop
    end if
end while

The forward selection procedure is applied separately to three model configurations: seasonal cells, non-seasonal cells, and the full dataset. Each configuration follows the same evaluation protocol, where features are added sequentially based on their impact on validation RMSE. The final selected feature set for each configuration is determined when no additional feature yields further improvement.

Table 7: Feature selection results across seasonal, non-seasonal, and all-cell models

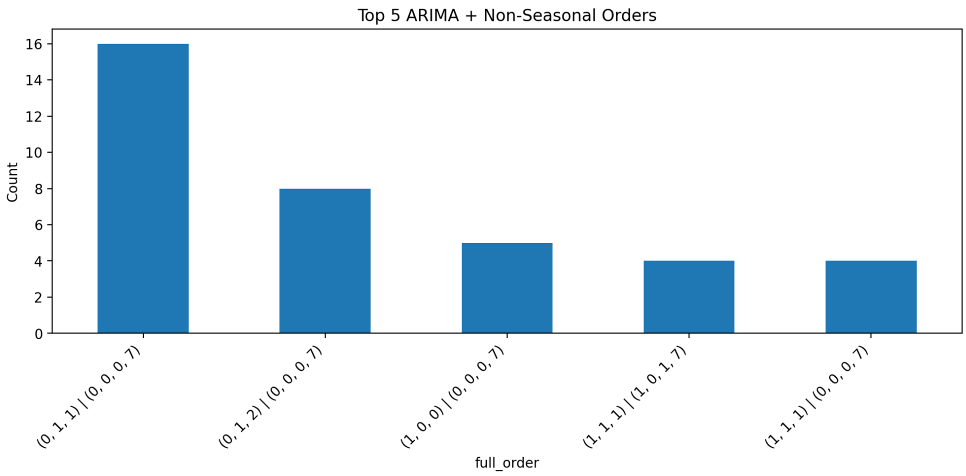
Model Type	Selected Features (final step)	RMSE
Seasonal cells	roll_mean_7, roll_max_7	1.62
Non-seasonal cells	roll_min_7, lag_1, is_weekend	1.71
All cells	roll_min_7, lag_1, lag_diff_1_7	1.66

Table 7 shows the results from the forward feature selection. A strong overlap exists between the non-seasonal and all-cell configurations, both selecting `roll_min_7` and `lag_1` as the most predictive features. The non-seasonal configuration additionally benefits from the `is_weekend` indicator, which is consistent with the commuter driven daily patterns observed in urban cells, where weekday and weekend traffic differ systematically. Seasonal cells, by contrast, rely more on short-term aggregated statistics such as rolling means and maxima, suggesting that recent traffic levels are more informative than specific calendar effects.

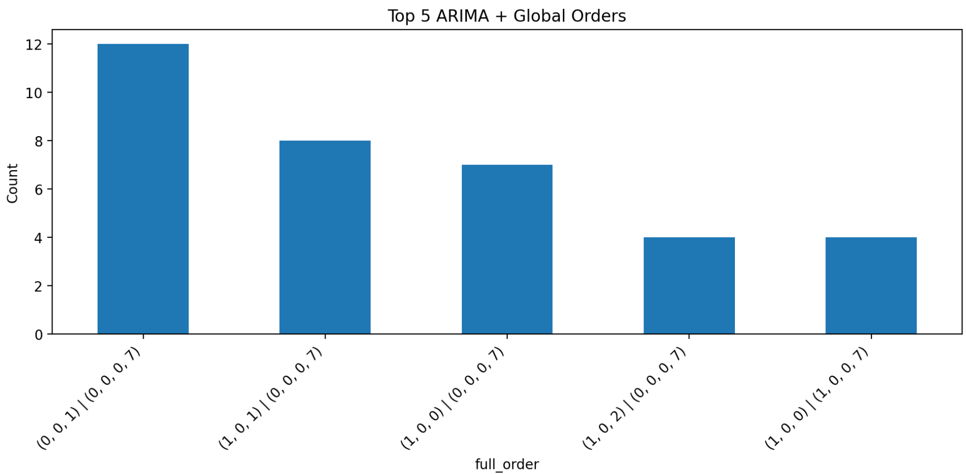
Post feature selection order refinement

After feature selection, the top ARIMA orders identified during the baseline stage were re-evaluated using the selected exogenous variables. Each configuration was refitted and tested on the held-out dataset, and performance was compared to the feature-selection baseline.

(a) ARIMA order frequency for seasonal cells



(b) ARIMA order frequency for non-seasonal cells



(c) ARIMA order frequency for all cells

Figure 14: ARIMA order selection after feature engineering

The results show only limited changes after feature selection. For seasonal cells, a slightly different configuration ARIMA$(0, 0, 1)(0, 0, 0, 7)$ achieved the best performance with an RMSE of 1.51. For non-seasonal cells and all cells, the original configuration

ARIMA$(0, 1, 1)(0, 0, 0, 7)$ remained optimal.

Table 8: Post feature selection ARIMA performance comparison

Model Type	Best Order	RMSE
Seasonal cells	$(0, 0, 1)(0, 0, 0, 7)$	1.5139
Non-seasonal cells	$(0, 1, 1)(0, 0, 0, 7)$	1.7102
All cells (global)	$(0, 1, 1)(0, 0, 0, 7)$	1.6600

Overall, the results suggest that both feature engineering and model order selection contribute to forecasting performance. Feature selection provides the main improvement, while re-tuning the ARIMA order after feature selection yields only marginal additional gains.

For seasonal cells, a small but consistent improvement is observed after adjusting the ARIMA order, reducing the RMSE from the feature-selection baseline. For non-seasonal and global models, changes in ARIMA order do not lead to meaningful improvements.

This indicates that exogenous features carry most of the predictive signal, while ARIMA order tuning provides secondary refinement in certain cases rather than large performance shifts.

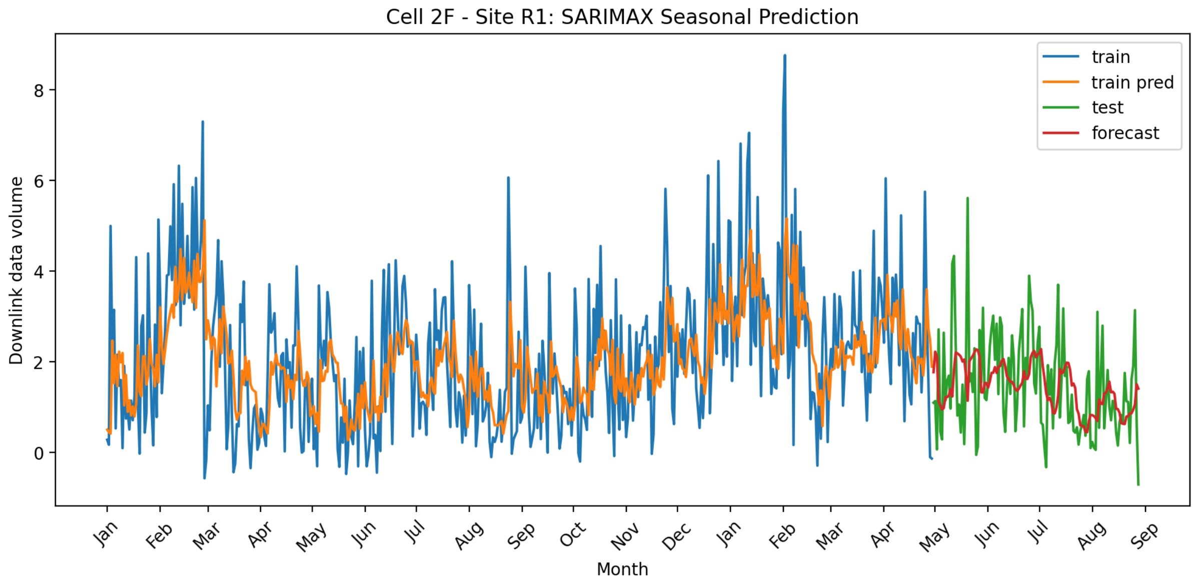


Figure 15: SARIMAX Forecasting on R1 2F

To better understand model behaviour, Figure 15 shows the fitted values on the training data and the predictions on the held-out test set for a representative cell (R1, 2F).

5.3 Ordinary Least Squares

Ordinary Least Squares (OLS) is a linear regression method that estimates parameters by minimizing the sum of squared errors between observed values and predictions [10].

In the simplest case with one feature, OLS fits a line of the form:

$$y = \beta_0 + \beta_1 x$$

where β_0 is the intercept and β_1 is the slope. These are chosen to minimize:

$$\min_{\beta_0,\beta_1} \sum_{i=1}^{n} (y_i - (\beta_0 + \beta_1 x_i))^2$$

Unlike ARIMA, which models temporal dependence using past values of the time series, OLS does not assume any time structure. Each observation is treated as independent. With multiple features, OLS generalizes to:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_p x_p$$

or in matrix form:

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$$

Here, $\mathbf{X}$ is the feature matrix, $\boldsymbol{\beta}$ is the vector of learned coefficients, and $\boldsymbol{\epsilon}$ is the error term.

Each feature contributes additively to the prediction through its own weight, along with a single shared intercept. This allows OLS to incorporate engineered features such as lagged values, rolling statistics, calender effects or external variables.

Feature selection

The same engineered feature pool and forward selection procedure used for the SARIMAX model were applied to the OLS model, as shown in Table 6 and Algorithm 1. The evaluation metric and stopping criterion were also kept identical, ensuring a fair comparison between the two approaches.

Table 9: Feature selection results across seasonal, non-seasonal, and all-cell models

Model Type	**Selected Features (final step)**	**RMSE**
Seasonal cells	roll_mean_7, lag1, roll_mean_14, roll_std_14, holiday	1.45
Non-seasonal cells	roll_mean_14, lag1, roll_std_14, is_weekend	1.60
All cells	roll_mean_7, lag1, roll_std_7, roll_mean_14, is_weekend	1.56

Compared to the ARIMA feature selection results, OLS selects a larger feature set across all configurations, incorporating additional rolling statistics and calendar variables such as `holiday` and `is_weekend`. Despite treating observations as independent, OLS achieves lower RMSE than ARIMA/SARIMAX across all groups (1.45 vs 1.51 for seasonal cells, 1.60 vs 1.71 for non-seasonal cells), suggesting that the additional features compensate for the lack of explicit time-series modelling. This indicates that when sufficiently rich features are provided, a simple linear model can match or outperform more complex temporal models.

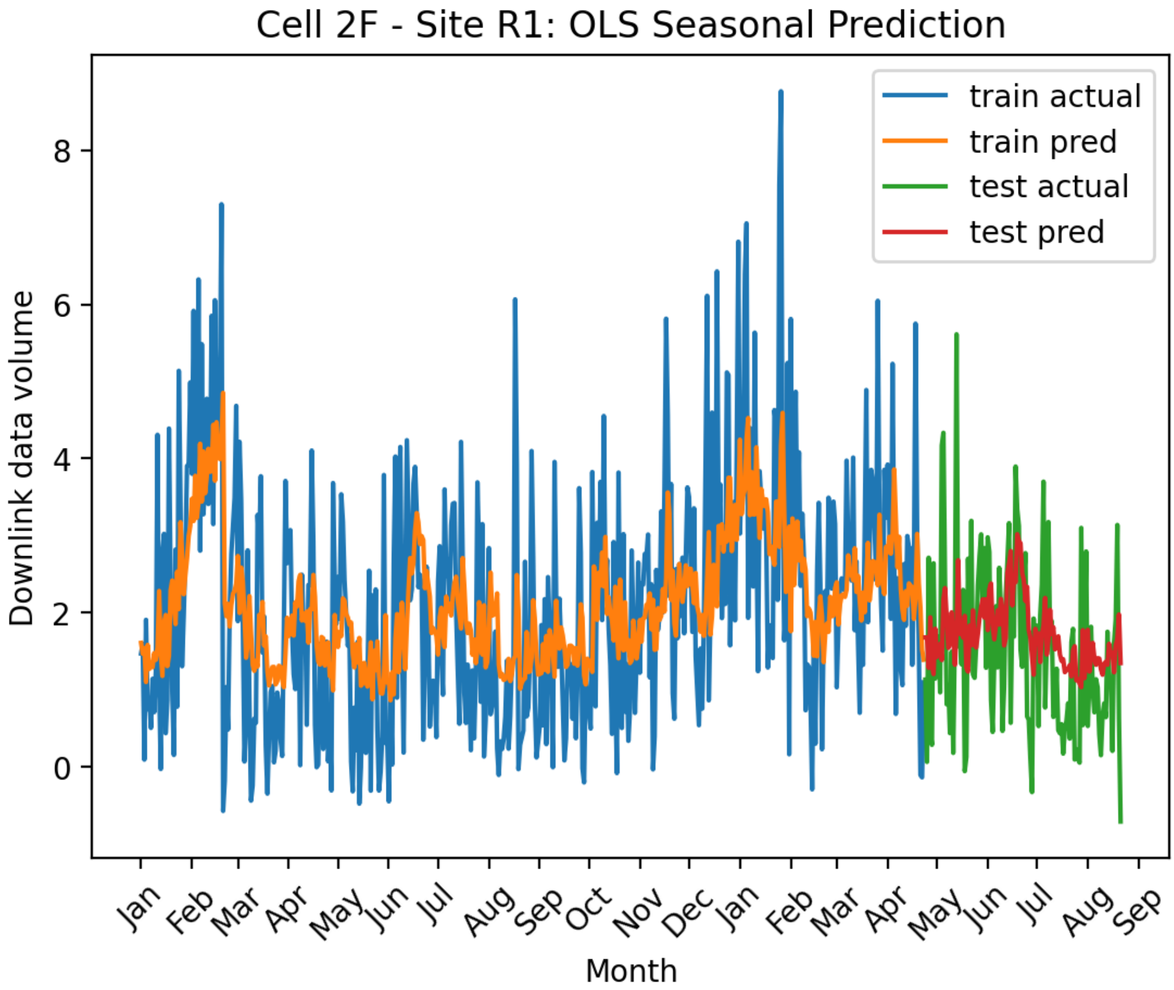


Figure 16: OLS forecasting on site R1 cell 2F

To better understand model behaviour, Figure 16 shows the OLS predictions on the cell used in the ARIMA evaluation (R1, 2F).

5.4 XGboost

Unlike the statistical models presented in the previous sections, XGBoost (Extreme Gradient Boosting) is a machine learning method that makes no parametric assumptions about the underlying data generating process [13]. Where ARIMA models temporal dependence explicitly through autoregressive and moving average components, and OLS assumes a fixed linear relationship between features and the target, XGBoost learns complex non-linear relationships directly from data by sequentially fitting an ensemble of decision trees, each correcting the residual errors of the previous one.

A key practical distinction from the models presented earlier is that XGBoost is data-hungry. Cell-specific models, as applied in the ARIMA and OLS sections, operate on a single time series at a time and are therefore well suited to small, isolated datasets. XGBoost however benefits substantially from volume and diversity of training examples. For this reason, all cells across all sites are pooled into a single dataset, allowing the model to learn shared patterns across the entire network of cells simultaneously.

Decision tree

A decision tree makes predictions by repeatedly applying simple threshold rules to the input features. Each internal node in the tree checks a condition of the form "is feature $x_j \leq t$?" where x is a feature and t is a threshold, which is equivalent to a sequence of nested if-else statements. Based on the result of each test, the observation is routed left or right down the tree.

This process continues until the observation reaches a terminal node (a leaf). Each leaf stores a fixed prediction value. The final model output is simply the value stored in the leaf where the observation ends up after following all the splits.

A decision tree does not learn a smooth function or a set of linear weights, but instead partitions the feature space into a set of rectangular regions, where all observations in the same region share the same predicted value. A single tree is therefore easy to interpret but can easily overfit if it becomes too deep.

XGBoost addresses this by combining many trees into an ensemble, but unlike approaches that train trees independently on bootstrapped subsets (e.g. random forests) and average their predictions, XGBoost builds trees sequentially. Each new tree is trained to correct the residual errors made by the previous ensemble, rather than being trained independently. The final prediction is the sum of all trees:

XGBoost follows the same idea of combining many trees, but instead of training them independently, it builds them sequentially. Each new tree is trained to correct the errors made by the previous ensemble. The final prediction is the sum of all trees:

$$\hat{y}_i = \sum_{k=1}^{K} f_k(\mathbf{x}_i)$$

This approach is known as gradient boosting, and XGBoost is a highly optimised implementation of it.

Implementation

Since XGBoost processes one observation at a time and has no built-in notion of time or sequential order, it cannot look back at past values the way ARIMA does. This makes feature engineering particularly important. All temporal context that the model needs to understand historical behaviour must be explicitly constructed and provided as input features. With this in mind, the feature set was designed to capture both temporal patterns and cell-specific characteristics, and can be grouped into the following categories.

Raw network metrics were taken directly from the network performance management (PM) counters. These KPIs include downlink and uplink traffic volume, PRB utilisation, active user count, throughput, and SINR measurements. They represent the observed state of each cell at each time step and are used as direct inputs to the models.

Lag features capture historical values of the target at fixed time offsets: 1, 8, 24, 168, and 720 hours back. These allow the model to condition its prediction on recent observations as well as patterns from the same hour yesterday, the same day last week, and the same period last month. **Rolling statistics** were computed across a wide range of windows: 3, 6, 12, 24, 48, 168, 336, 720, and 2160 hours. For each window, the mean, maximum, minimum, and standard deviation were included. This gives the model a rich view of both short-term fluctuations and long-term trends.

Calendar features were one-hot encoded to allow the model to learn non-linear patterns tied to time. This includes the hour of the day (0–23), day of the week (0–6), and month of the year (1–12), as well as binary indicators for weekends and public holidays.
Cell and technology features were included to allow the model to differentiate between cells. The site group and technology generation (4G/5G) were one-hot encoded, along with the frequency band, covering LTE800, LTE1800, LTE2600, NR700, and NR3500. A binary seasonal indicator was also included based on the pre-classification described in earlier sections.

In total this results in **107 input features** per observation.

XG Boost Results

The model was trained with 200 trees, a learning rate of 0.05, and a maximum tree depth of 4. The data was split chronologically, with the first 80% of observations used for training and the remaining 20% reserved for testing. Table 10 summarizes the forecasting performance across the different dataset splits.

Table 10: XGBoost forecasting performance

Dataset	RMSE
Seasonal cells	1.46
Non-seasonal cells	1.56
All cells	1.53

XGBoost achieves an RMSE of 1.46 on seasonal cells, 1.56 on non-seasonal cells, and 1.53 across all cells combined, despite being trained on all cells pooled together rather than per cell.

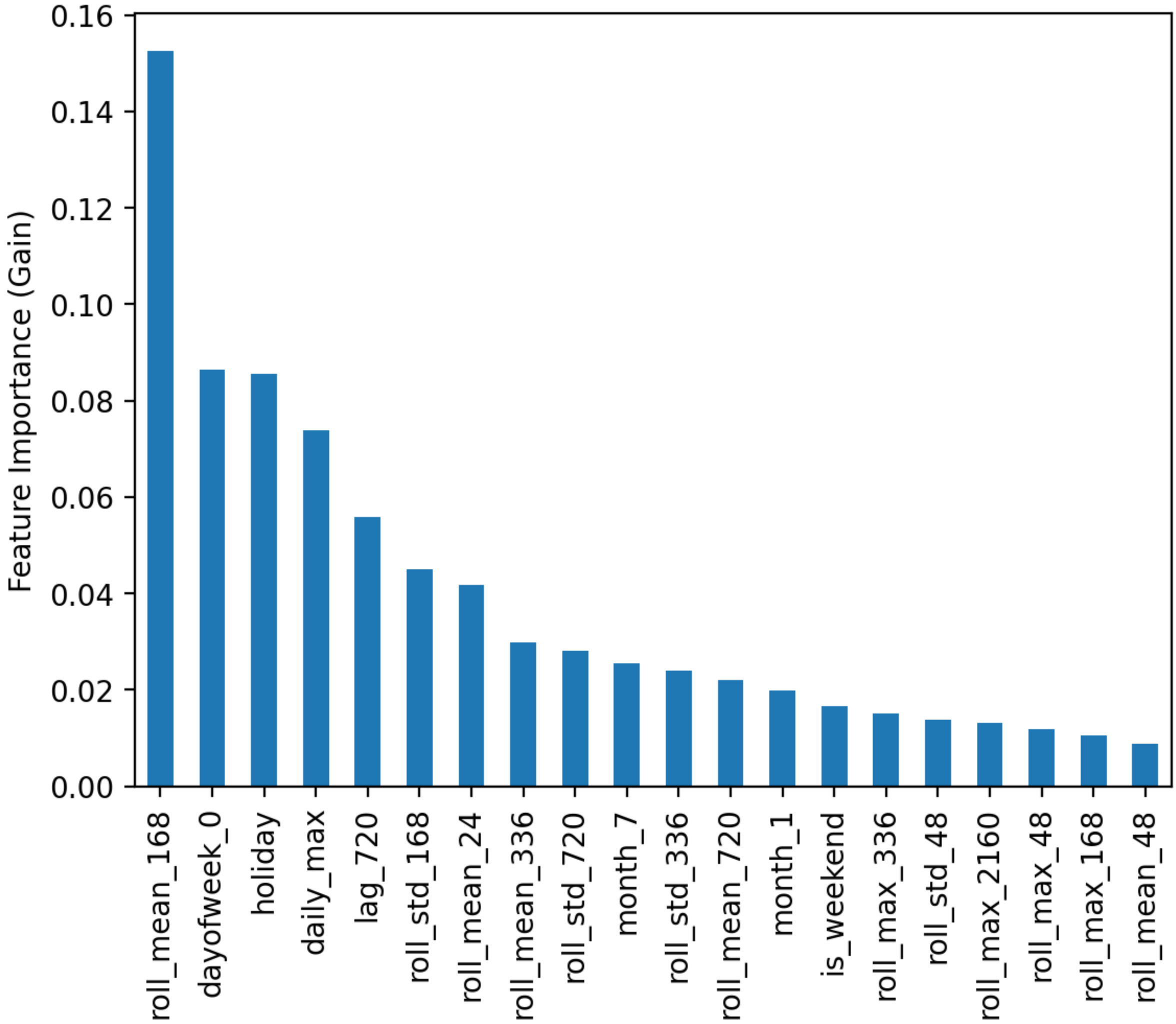


Figure 17: XG Boost feature importance

Figure 17 shows the feature importances assigned by the model, measured by gain, which reflects how much each feature reduces the prediction error on average across all splits where it is used. The most influential feature was `roll_mean_168`, the rolling mean over the past week, with a gain of 0.15. This suggests that the average traffic level over the previous seven days is the single strongest predictor of the next day daily maximum. Calendar features also rank highly, with `dayofweek_0` and `holiday` achieving gains of 0.09 and 0.09 respectively, indicating that the model has picked up on weekly rhythms and public holiday effects. Other notable features include `daily_max` at 0.07, `lag_720` at 0.06, and `roll_std_168` at 0.04, reinforcing that medium to long-term historical behaviour dominates the predictions.

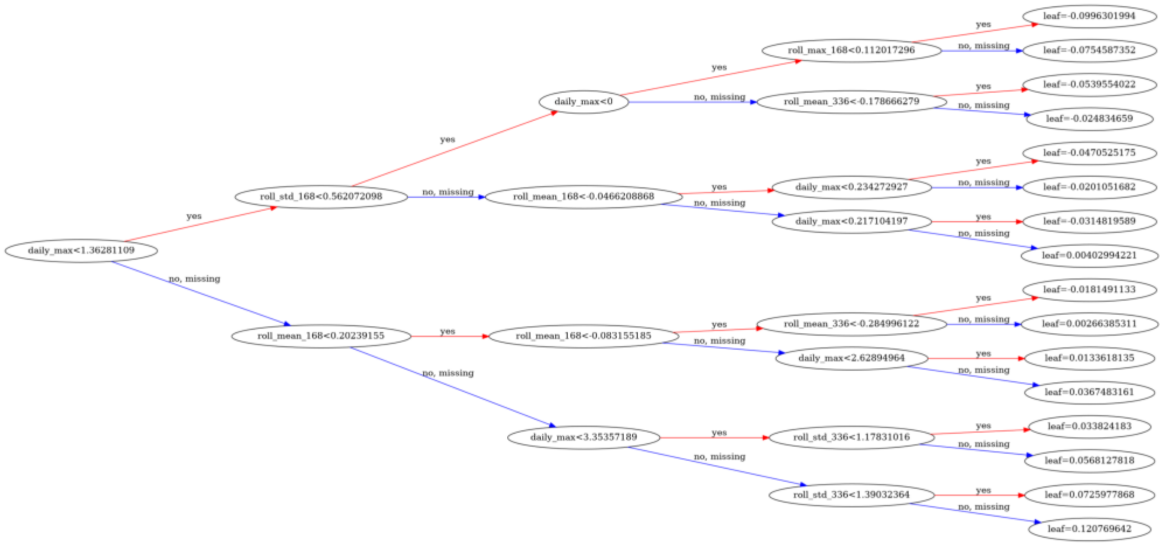


Figure 18: A single tree from XG model

Figure 18 visualises a single tree from the ensemble. At each internal node the tree makes a binary decision based on a threshold for a given feature, routing each observation either left or right until it reaches a leaf node where a prediction is made. With a maximum depth of 4, the tree can make at most 4 consecutive splits, resulting in up to 16 leaf nodes each outputting a distinct predicted value.

The root node splits on `daily_max`, by far the highest-gain split in the tree, and subsequent splits rely mainly on rolling mean and standard deviation features rather than calendar variables. While this is only one of 200 trees in the model, it is illustrative of the general pattern seen in the feature importance results 17, where recent traffic level dominate over calendar-based features.

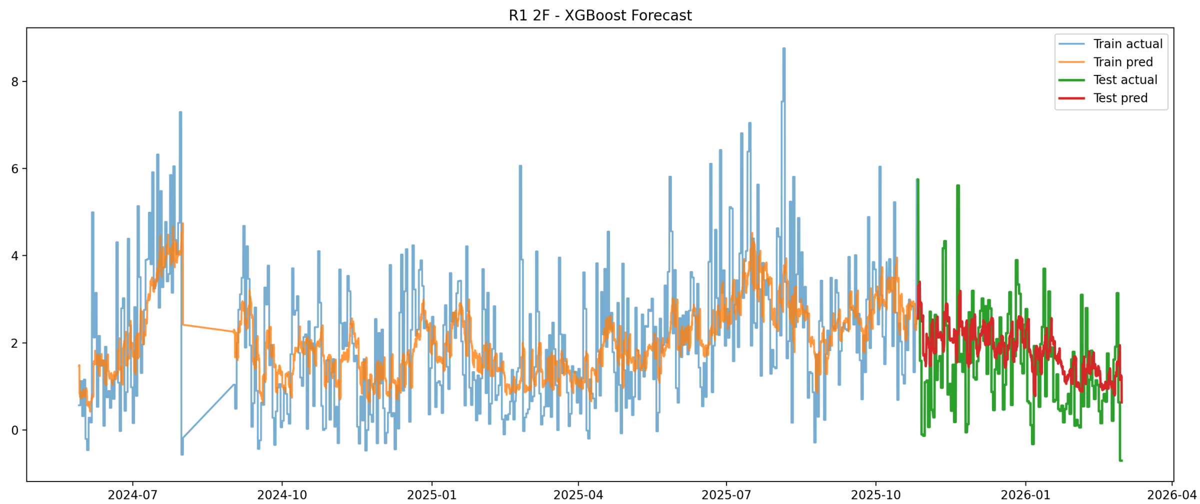


Figure 19: Prediction vs Actual, xg boost on site R1 cell 2F

Figure 19 shows the predicted versus actual daily maximum for site R1, cell 2F. The model captures the general trend and seasonal fluctuations reasonably well, though some peaks are underestimated, which is a common characteristic of tree based models that predict based on previously seen value ranges.

6 Results and Discussion

This section presents the results of the forecasting models. First, forecasting performance is compared across all models. Then the impact of feature engineering and dataset splitting is evaluated.

6.1 Model Comparison

To ensure a fair comparison across all modelling approaches, RMSE is computed individually for each cell and the average RMSE is then taken across all cells within a given group. This applies consistently to the naive baselines, ARIMA/SARIMAX, OLS, and XGBoost, ensuring that no model is favoured by aggregating residuals differently or allowing high-traffic cells to disproportionately dominate the metric.

Table 11 summarises the performance of all forecasting approaches across seasonal, non-seasonal, and combined datasets. The results provide a complete comparison between naive baselines, statistical models, and machine learning approaches, allowing an assessment of both feature engineering and model complexity.

Table 11: Final model comparison across forecasting approaches

Model	**Seasonal**	**Non-seasonal**	**All cells**
Seasonal Naive	2.36	2.58	2.51
Mean Naive	1.85	1.93	1.93
Previous Naive	1.77	1.99	1.92
ARIMA/SARIMAX	1.51	1.71	1.66
OLS	1.45	1.60	1.56
XGBoost	1.46	1.56	1.53

A clear pattern emerges when comparing the naive baseline methods with the more advanced forecasting models. All model-based approaches outperform the baseline methods across seasonal, non-seasonal, and combined datasets. While the best baseline models achieve RMSE values between 1.77 and 2.58, the statistical and machine learning models reduce this to 1.45–1.71, demonstrating that the dataset contains substantial and predictable temporal structure.

A more nuanced picture emerges once XGBoost is compared against the per-cell statistical models. XGBoost achieves the lowest RMSE among all models for non-seasonal cells (1.56) and across all cells combined (1.53), and performs comparably to OLS on seasonal cells (1.46 vs 1.45).

OLS remains highly competitive and achieves the best result on seasonal cells (1.45), narrowly ahead of XGBoost (1.46) and meaningfully ahead of ARIMA (1.51). For non-seasonal and all-cell settings.

ARIMA/SARIMAX consistently performs worst with RMSE values of 1.51, 1.71, and 1.66 for seasonal, non-seasonal, and all cells respectively.

Overall, the comparison indicates that feature engineering is a major driver of predictive performance across all models, while the relative ranking between OLS and XGBoost depends on the dataset split. The following section formally tests whether these observed differences are statistically significant.

6.2 Statistical Significance of Model Differences

To assess whether observed differences in RMSE are statistically meaningful, paired t-tests were conducted on the per-cell RMSE values, treating each cell as one paired observation across seasonal, non-seasonal, and all cells. In addition, bootstrap confidence intervals were computed for the mean difference in RMSE.
Table 12 summarises the results for seasonal cells.

Table 12: Statistical comparison of forecasting models across dataset splits

Comparison	**Mean diff**	**95% CI**	**t-value**	**p-value**
Seasonal cells				
ARIMA - OLS	0.062	[0.029, 0.098]	3.49	0.0011
XGBoost - OLS	0.008	[-0.018, 0.029]	0.81	0.4200
XGBoost - ARIMA	-0.053	[-0.08, -0.027]	3.91	0.0003
Non-seasonal cells				
ARIMA - OLS	0.107	[0.066, 0.153]	4.62	0.000013
XGBoost - OLS	-0.040	[-0.059, -0.021]	4.07	0.000099
XGBoost - ARIMA	-0.148	[-0.197, -0.101]	5.88	0.000001
All cells				
ARIMA - OLS	0.104	[0.070, 0.142]	5.76	< 0.001
XGBoost - OLS	-0.025	[-0.040, -0.010]	3.34	0.0011
XGBoost - ARIMA	-0.129	[-0.169, -0.093]	6.73	< 0.001

The seasonal subset is the only split where the ranking from Table 11 does not hold up statistically. While OLS (1.45) and XGBoost (1.46) appeared nearly identical, the test confirms this: the confidence interval for their difference spans zero ([-0.018, 0.029], $p = 0.42$), so there is no evidence the two models differ at all on seasonal cells. ARIMA, by contrast, is significantly worse than both. Its gap with OLS (0.062, $p = 0.0011$) and with XGBoost (0.053, $p = 0.0003$) confirming ARIMA's weaker performance on seasonal cells is a real effect rather than chance.

For non-seasonal cells, every comparison is significant, but the effect sizes differ considerably. ARIMA's gap with OLS is significant at 0.107 ($p < 0.001$), meaning ARIMA's underperformance is most pronounced on non-seasonal cells. XGBoost's advantage over OLS is smaller but still clearly significant at 0.040 ($p < 0.001$), with a confidence interval ([-0.059, -0.021]) that stays comfortably below zero, so unlike the seasonal case, this is a real, if modest, edge for XGBoost. Combined, ARIMA trails XGBoost by 0.148 ($p < 0.001$), the largest effect size in the entire table.

The all-cells results mirror the non-seasonal pattern rather than the seasonal one, which makes sense given non-seasonal cells make up the majority of the dataset. XGBoost retains a significant edge over OLS (0.025, $p = 0.0011$), and both significantly outperform ARIMA.

Overall, the testing changes the conclusion in one important way, the apparent OLS advantage on seasonal cells in Table 11 is not statistically supported, OLS and XGBoost should be treated as equivalent there. Everywhere else, the gaps are real, ARIMA is reliably the weakest model in every split, and XGBoost holds a genuine, if small, edge over OLS specifically on non-seasonal and combined data.

6.3 Impact of seasonal pre classification

To further investigate the role of preprocessing choices, an ablation study was conducted comparing global models trained on all cells with models trained on group-specific subsets (seasonal and non-seasonal). This allows evaluation of whether dataset segmentation provides additional predictive benefit beyond feature engineering alone.

Table 13 summarises the results.

Table 13: Effect of global vs group-specific ARIMA and OLS models

	ARIMA		**OLS**	
Training Strategy	**Seasonal**	**Non-seasonal**	**Seasonal**	**Non-seasonal**
Global model	1.53	1.72	1.47	1.60
Seasonal-specific model	1.51	N/A	1.45	N/A
Non-seasonal-specific model	N/A	1.71	N/A	1.60

The results show that dataset segmentation has only a limited impact on performance. For ARIMA, a small improvement is observed for seasonal cells, where RMSE decreases from 1.53 to 1.51 (approximately 1.3%). For non-seasonal cells, the difference between global and specialised models is negligible (1.72 vs 1.71), corresponding to an improvement of around 0.6%.

For OLS, a small improvement is observed for seasonal cells, where RMSE decreases from 1.47 to 1.45 (approximately 1.4%) when using a group-specific model. For non-seasonal cells, performance remains unchanged at 1.60, indicating that partitioning provides no additional benefit for this group.

The lack of improvement for non-seasonal cells can be explained by the imbalance in the dataset, where non-seasonal cells represent approximately 69% of all samples (96 vs 43 seasonal cells). As a result, the global model is dominated by non-seasonal behaviour and already closely approximates this subgroup, reducing the benefit of training a separate model.

outperform naive baselines, confirming that the dataset contains strong and learnable temporal structure.

Splitting the dataset into seasonal and non-seasonal groups provides seem to only provide marginal improvements overall. However, the largest gains are consistently observed for the seasonal subset, which represents the smaller minority group (approximately 31% of all cells). This suggests that group-specific models are most beneficial when the target subset is underrepresented in the global training data, as the global model is otherwise dominated by the majority class behaviour. For the non-seasonal subset, which already represents 69% of all samples, group-specific training provides little to no additional benefit.

Overall, these results indicate that while segmentation can provide small but consistent gains for minority groups, the majority of the performance is already achieved through global modelling combined with feature engineering. The added complexity of maintaining separate pipelines is therefore only justified when a meaningful class imbalance exists between seasonal and non-seasonal cells, which does not appear to be the case in this dataset.

7 Conclusion

This thesis investigated whether pre-classifying mobile network cells based on traffic behaviour can improve forecasting performance. A statistical classification method was developed using a t-test on holiday and non-holiday traffic differences, separating cells into seasonal and non-seasonal groups. Forecasting models including ARIMA/SARIMAX, OLS, and XGBoost were then evaluated on these groups.

The classification results show clear structure across geographic site types. Urban cells are consistently classified as non-seasonal (52/52), while recreational and suburban cells show more mixed behaviour. Overall, 88 out of 91 cells in urban and recreational categories align with their expected classification based on geographic structure. Importantly, this separation was achieved purely from traffic behaviour, without using geographical location as an input, demonstrating that the method captures meaningful patterns in a data-driven and location-agnostic manner.

It should be noted that this study used geographical location as a proxy for ground truth when evaluating the classification results. However, geographic category does not necessarily determine traffic behaviour, and the two may diverge in practice. A clear example is R3, which despite being geographically labeled as a recreational site, shows low holiday sensitivity similar to urban cells (Figure 11). Conversely, U3, despite being geographically labeled as urban, exhibits a daily activity pattern closer to those of recreational cells (Figure 8), with low daytime activity and a pronounced evening peak. This suggests that that relying solely on location-based grouping may introduce misclassification. In the absence of true ground truth labels at the cell level, the agreement between the classification and geographic categories should therefore be interpreted as supporting evidence rather than definitive validation.

In the forecasting task, all model-based approaches significantly outperform naive baselines, with error reductions of approximately 19% compared to the best naive model. Best-performing models achieve RMSE values in the range of 1.45–1.70 across all datasets, compared to naive baseline RMSE values of 1.77–2.58. This confirms that the dataset contains strong temporal structure that can be learned using statistical and machine learning methods.

Among all models, OLS and XGBoost achieve comparable performance overall, while ARIMA/SARIMAX is consistently the weakest. Paired statistical testing across cells confirms that ARIMA performs significantly worse than both OLS and XGBoost in every dataset split. The relationship between OLS and XGBoost depends on the subgroup: the two models are statistically indistinguishable on seasonal cells (RMSE of 1.45 and 1.46 respectively), while XGBoost holds a small but statistically significant advantage over OLS on non-seasonal cells (1.56 vs 1.60) and across all cells combined (1.53 vs 1.56). Although this advantage is statistically significant, the absolute difference remains small, and OLS achieves this performance using a simple, interpretable linear model rather than an ensemble of decision trees. This suggests that while a complex, non-linear model such as XGBoost can extract a modest additional benefit from pooling data across cells, a well engineered linear model captures nearly all of the same predictive signal while remaining considerably more explainable and easier to deploy and maintain in practice. Overall, this indicates that strong feature engineering, particularly lagged values and rolling statistics, captures most of the predictive signal regardless of model choice, and that the added complexity of a model like XGBoost is only weakly justified by the resulting gains in this setting.

The impact of dataset segmentation is limited. Training separate models for seasonal and non-seasonal cells in the statistical models only results in marginal improvements compared to a global model, and only for the seasonal subset, which represents the smaller minority group. When a subgroup is already well represented in the global training data, as is the case for non-seasonal cells at 69% of all samples, group-specific training provides little to no additional benefit. This suggests that pre-classification is most valuable when a meaningful class imbalance exists, which does not appear to be the case in this dataset.

Overall, the results show that a simple pipeline combining statistical classification and feature-driven forecasting models, whether linear or tree-based, is sufficient to achieve strong performance in this setting when comparing to baseline models. Feature engineering plays a more important role than model complexity or segmentation of the data, although model choice still produces small, statistically significant differences depending on the seasonality of the subgroup.

Future work could extend the classification approach beyond a simple seasonal versus non-seasonal split. For example, additional categories could be introduced for event-driven cells, or cells could be scored on a continuous scale of seasonality rather than sorted into two fixed groups. Another direction could be classifying cells by the dominant device type or usage behaviour driving their traffic, such as distinguishing cells dominated by smartphone usage from those with a higher share of smart TV or streaming traffic, since these are likely to produce distinct volume and timing patterns. Unsupervised methods could be explored as an alternative to the t-test used in this thesis for identifying traffic regimes. Finally, deep learning models such as LSTMs or CNNs could be tested, as they may capture temporal and spatial patterns across cells more effectively than the linear and tree-based models used here.

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